

Computer Network

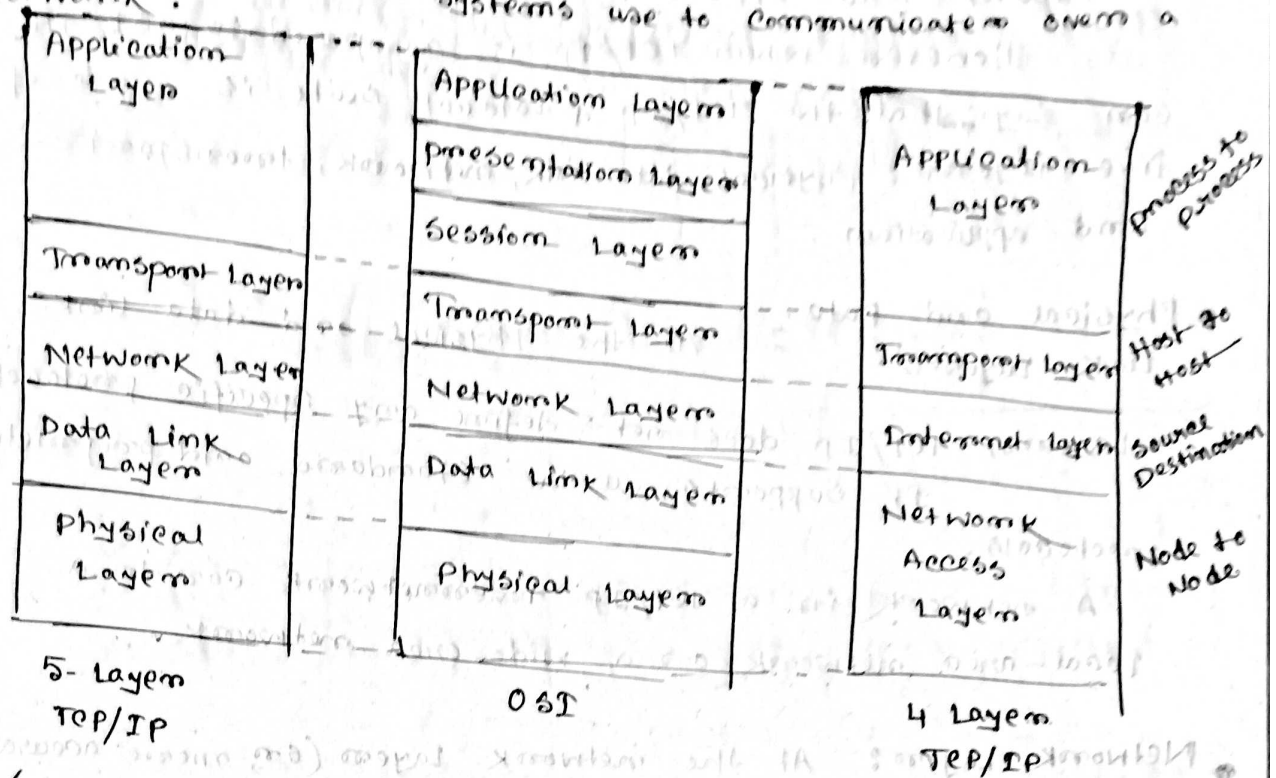
A Computer network is a collection of various computing devices. The purpose of computer network is so that the devices can share data.

	PAN	LAN	CAN	MAN	WAN
Full Form	personal Area Network	Local Area Network	Campus Area Network	Metropolitan Area Network	Wide Area Network
Technology	Bluetooth, IrDA, Zigbee	Ethernet and wifi	Ethernet	FDDI, CDDI, ATM	200 Leased Line, Dial-up
Range	1-100 Meter	upto 2 km	1-5 km	5-50 km	Above 50 km
Transmission speed	very High	very High	High	Average	Low
Area	Within a Room	Within office, building	Within university, Corporate offices	Within city like Mumbai	Within Countries
Ownership	private	private	private	private or public	private or public
Maintenance	Very Easy	Easy	Moderate	Difficult	Very difficult
Emmion Rate & Cost	Very Low	Low	Moderate	High	Very High

These are differentiate by the distances.

TCP/ IP protocol suite or Internet protocol

The open system interconnection (OSI) Model describes seven layers that computer systems use to communicate over a network.



Summary of layers in OSI Model

- Application: To allow access to network resources.
- Presentation: To translate, encrypt and compress data.
- Session: To establish, manage and terminate sessions.
- Transport: To provide reliable process-to-process message delivery and error recovery.
- Network: To move packets from source to destination, to provide internetworking.
- Data Link: To organize bits into frames. To provide hop-to-hop delivery. The main function of this layer is to make sure data transfer is error-free from one node to another.
- Physical: To transmit bits over a medium, to provide mechanical and electrical specifications.

TCP/IP protocol suite: (Internet protocol)

The layers in the TCP/IP protocol suite do not exactly match those in the OSI model.

The original TCP/IP protocol suite was defined as having four layers: host-to-network, internet, transport and application.

However when TCP/IP is compared to OSI, we can say that the TCP/IP protocol suite is made of five layers: Physical, data link, network, transport and application.

● Physical and Data Link Layers : At the physical and data link

layers, TCP/IP does not define any specific protocol. It supports all the standard and proprietary protocols.

A network in a TCP/IP internetwork can be a local-area network or a wide-area network.

● Network Layer : At the network layer (or, more accurately, the internetwork layer), TCP/IP supports the Internetworking protocol. IP, in turn, uses four supporting protocols: ARP, RARP, ICMP and IGMP.

ARP : Address Resolution protocol (ARP) is used to associate a logical address with a physical address. On a typical physical network, such as a LAN, each device on a link is identified by a physical or station address, usually imprinted on the network interface card (NIC). ARP is used to find the physical address of the node when its Internet address is known.

RARP : The Reverse Address Resolution protocol (RARP) allows a host to discover its Internet address when it knows only its physical address. It is used when a computer is connected to a network for the first time or when a diskless computer is booted.

ICMP : The Internet Control Message protocol (ICMP) is a mechanism used by hosts and gateway to send notification of datagram problems back to the sender. ICMP sends query and error reporting messages.

IGMP: The Internet Group Message protocol (IGMP) is used to facilitate the simultaneous transmission of a message to a group of recipients.

Transport Layer

Traditionally the transport layer was represented in TCP/IP by two protocols: TCP and UDP.

IP is a host-to-host protocol, meaning that it can deliver a packet from one physical device to another.

UDP and TCP are transport level protocols responsible for delivery of a message from a process (running program) to another process.

A few transport layer protocols, SCTP, has been devised to meet the needs of some newer applications.

UDP: The User Datagram protocol (UDP) is the simpler of the two standard TCP/IP transport protocols.

It is a process-to-process protocol that adds only port addresses, checksum error control, and length information to the data from the upper layer.

TCP: The Transmission Control protocol (TCP) provides full transport-layer services to applications.

TCP is a reliable stream transport protocol. The term stream, in this context, means connection-oriented: A connection must be established between both ends of a transmission before either can transmit data.

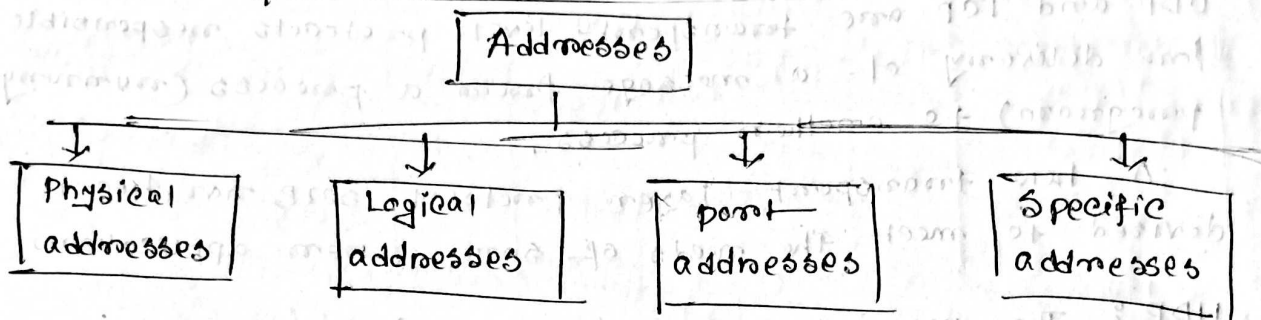
SCTP: The Stream Control Transmission protocol (SCTP) provides support for newer applications such as voice over the Internet. It is a transport layer protocol that combines the best features of UDP and TCP.

• Application Layer

The application layer in TCP/IP is equivalent to the combined session, presentation and application layers in the OSI model.

Many protocols are defined at this layer such as HTTP, FTP, SMTP etc.

Addresses in TCP/IP



Physical Address : The physical address, also known as the link address, is the address of a node as defined by its LAN or WAN.

- It is included in the frame used by the data link layer. It is the lowest-level address.

- The size and format of these addresses vary depending on the network. For example, Ethernet uses a 6 byte (48 bit) physical address that is imprinted on the network interface card (NIC). LocalTalk (Apple), however, has a 1-byte dynamic address that changes each time the station comes up.

07 : 01 : 02 : 01 : 2C : 4B

A 6-byte (12 hexadecimal digits) physical address.

Logical Address : Logical addresses are necessary for universal communications that are independent of underlying physical networks.

Physical addresses are not adequate in an internetwork environment where different

networks can have different address formats.

A universal addressing system is needed in which each host can be identified uniquely, regardless of the underlying physical network. The logical addresses are designed for this purpose.

A logical address in the Internet is currently a 32-bit address that can uniquely define a host connected to the Internet. No two publicly addressed and visible hosts on the Internet can have the same IP address.

There are 2 different versions of IP as follows —

1. IPv4 (IP version 4) :

IPv4 employs a 32-bit address. It is composed of four numbers separated by a 'dot' i.e., periods called an octet (byte). Each number in the octet can range from 0 to 255.

Example : 172.166.3.28

2. IPv6 (IP version 6) :

IPv6 is the next generation of Internet protocol addresses. In comparison to IPv4, IPv6 has a larger address space. IPv6 has a length of 128 bits and is written in hexadecimal. It is composed of eight fields, each of which contains two octets. As a result, IPv6 has 16 octets in total.

Example : 3221:1cd7:74b6:ebda7:0000:0006;7349:6442

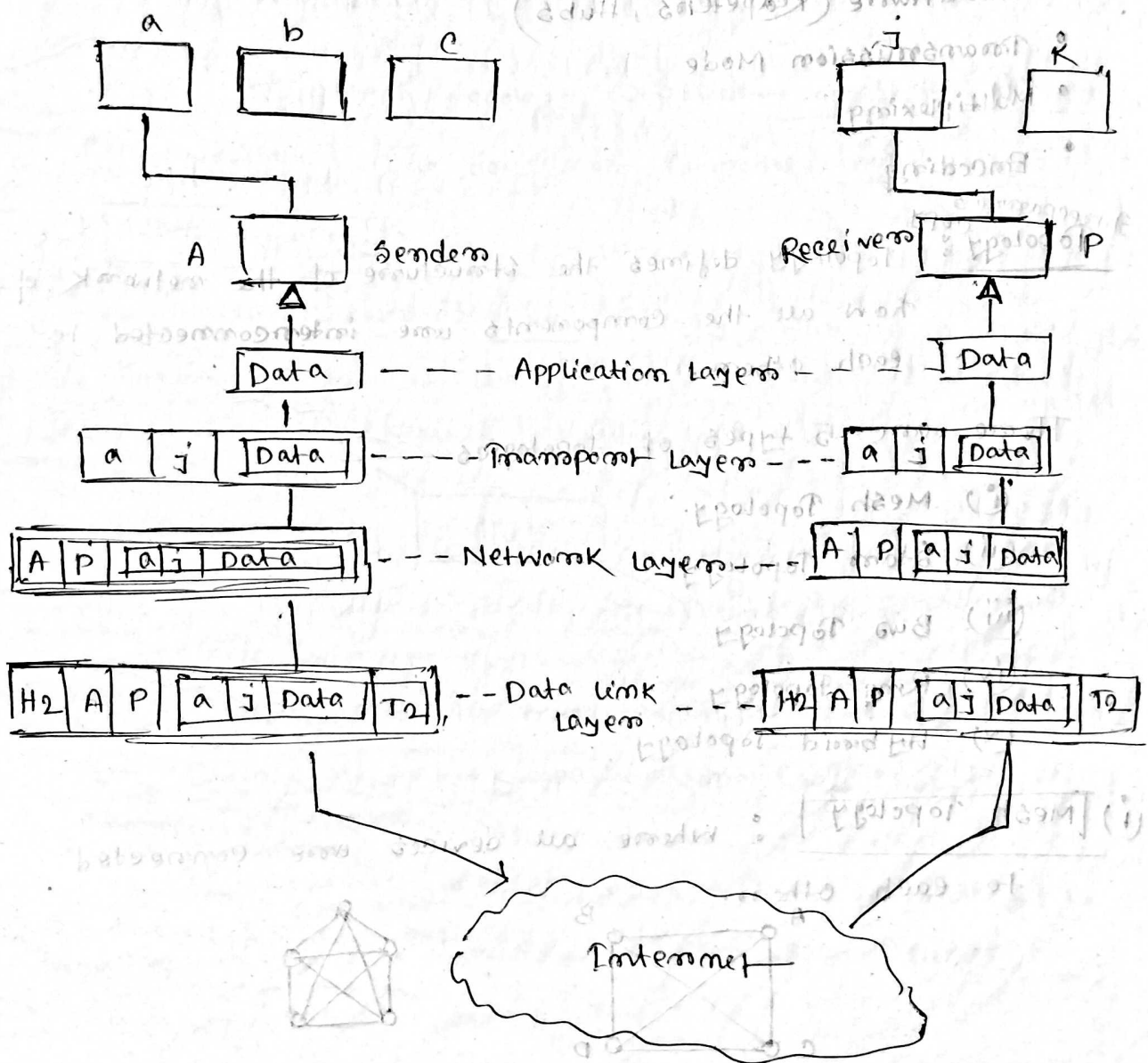
MAC ADDRESS	IP ADDRESS
(i) A unique identifier assigned to a network Interface Controller (NIC) of the computing device.	(i) A numerical label assigned to each device connected to a computer network that uses the Internet protocol for communication.
(ii) Stands for Media Access Control Address.	(ii) Stands for Internet protocol Address.
(iii) Also called physical, hardware or ethernet address.	(iii) Also called logical address, network or internet address.
(iv) Helps to uniquely identify the device.	(iv) Helps to identify the connection of a device on the internet.
(v) Assigned by the device manufacture.	(v) Assigned by an administrator or an ISP.
(vi) cannot be changed.	(vi) can be changed.
(vii) 48 bits (6 bytes) long.	(vii) IPV4 - 32 bits (4 bytes) IPV6 - 128 bits (16 bytes)
(viii) works in the data link layer.	(viii) works in the network layer.

Point

Address : The IP address and the physical address are necessary for a quantity of data to travel from a source to the destination host. However, arrival at the destination host is not the final objective of data communications on the Internet. A system that sends nothing but data from one computer to another is not complete.

Today, computers are devices that can run multiple processes at the same time. The end objective of Internet communication is a process communicating with another process.

For example, Computer A can communicate with Computer C by using TELNET. At the same time, Computer A communicates with Computer B by using the File Transfer protocol (FTP). For these processes to receive data simultaneously, we need a method to label the different processes. In other words, they need addresses. In the TCP/IP architecture, the label assigned to a process is called a port address. A port address in TCP/IP is 16 bits in length.



Specific Address :

Some applications have user-friendly addresses that are designed from that specific address.

Examples include the e-mail address (for example, `forouzam@fhda.edu`) and the universal Resource Locator (URL) (for example, `www.mtthe.com`).

The physical addresses change from hop to hop but the logical and port address usually remain the same.

Physical Layer and its functionalities

- Cables and Connectors
- Physical topology.
- Hardware (Repeaters, Hubs)
- Transmission Mode
- Multiplexing
- Encoding

Shreepanna's Copy Topology :

Topology defines the structure of the network how all the components are interconnected to each other.

There are 5 types of topologies —

(i) Mesh Topology

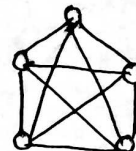
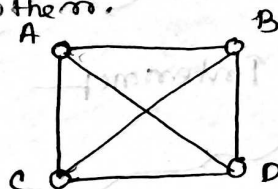
(ii) Star Topology

(iii) Bus Topology

(iv) Ring Topology

(v) Hybrid Topology

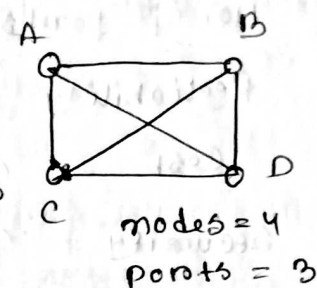
(i) Mesh Topology : Where all devices are connected to each other.



No. of Cables : $\frac{n(n-1)}{2}$ where $n = \text{no. of nodes}$.

No. of ports : $(n-1) * n$

Reliability : The mesh topology networks are very reliable as if any link breakdown will not affect the communication between connected computers.



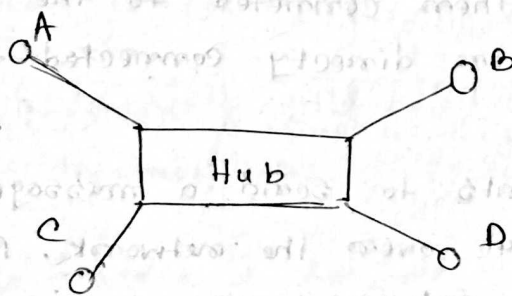
Cost : A mesh topology contains a large no. of connected devices such as a routers and more transmission media than other topologies.

Security : If 'A' is sending message to D. Then B and C will not even know. So B and C will not have any information. That is what it provides the security.

It Mesh topology, support point to point communication but does not support multi-casting.

(ii) Star topology:

Also known as hub topology. Hub is a multi-port device.



So, here multiple device are connected through hub.

~~Then~~ They are not directly connected but are connected through hub.

If A sends message to C then it will go to C through the hub.

No. of Cables = n (If n no. of devices then no. of cable required is (n))

No. of points = $1 \times n = n$

Reliability = Reliability is not there.

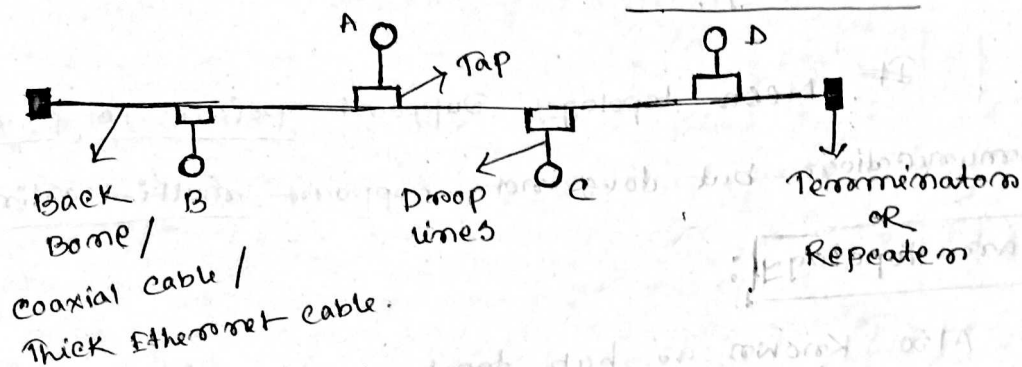
Cost = Cost is less as compared to mesh.

Security = Security is also there because if A is sending message to C then B and D will not know.

It works in broadcast way.

This is also support the point-to-point communication.

(iii) Bus topology : The bus topology is designed in such a way that all the stations are connected through a single cable known as a backbone cable.



Each node is either connected to the backbone cable by drop cable or directly connected to the backbone cable.

When a node wants to send a message over the network it puts a message over the network. All the stations available in the network will receive the message whether it has been addressed or not.

No. of Cables = $n+1$ Cost = Cost is less as compared to mesh topology.

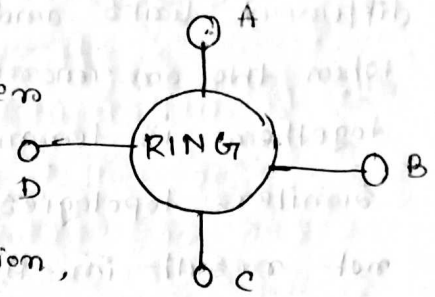
No. of points = n

Reliability = No reliability is there.

Security = If A is sending a message to B so 'C' and 'D' will know. So there is no security.

This is multipoint network.

(iv) **Ring Topology**: Ring topology is like a bus topology, but with connected ends. The node that receives the message from the previous computer will retransmit to the next node.



- The data flows in one direction, i.e., it is unidirectional.

- The data flows in a single loop continuously known as an endless loop.

- It has no terminated ends, i.e., each node is connected to other node and having no termination point.

- The data in a ring topology flow in a clockwise direction.

- The most common access method of the ring topology is token passing.

→ Token passing: It is a network access method in which token is passed from one node to another node.

→ Token: It is a frame that circulated around the network.

No. of cables = $n+1$

No. of points = n

Reliability = less

Security = less

Cost = cheap

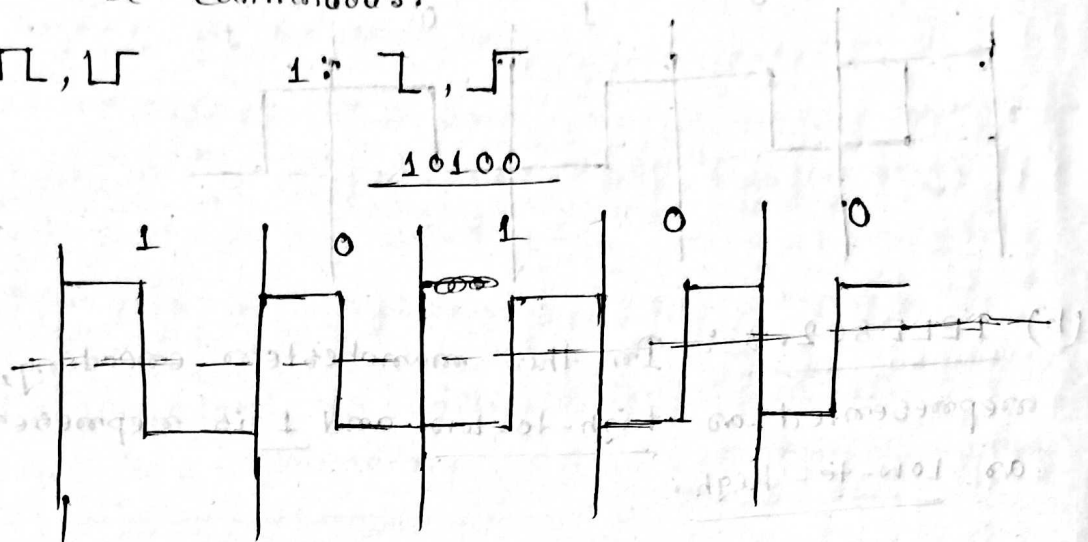
NOTE

Differential Manchester Encoding :

- It is also known as the Biphase mark code, etc.
- The presence and absence of the transition indicate the value.
- In differential Manchester Encoding 0 should contain an edge but 1 should not contain any edge it should be continuous.

0: $\sqcap, \sqcup$

1: $\sqsubset, \sqsupset$



Application on Differential Manchester Encoding :

- For every bit, there is a transition guaranteed.
- used in 802.5 with Twisted pair.

H.W.:

① 1000011

② 000000

③ 1111111

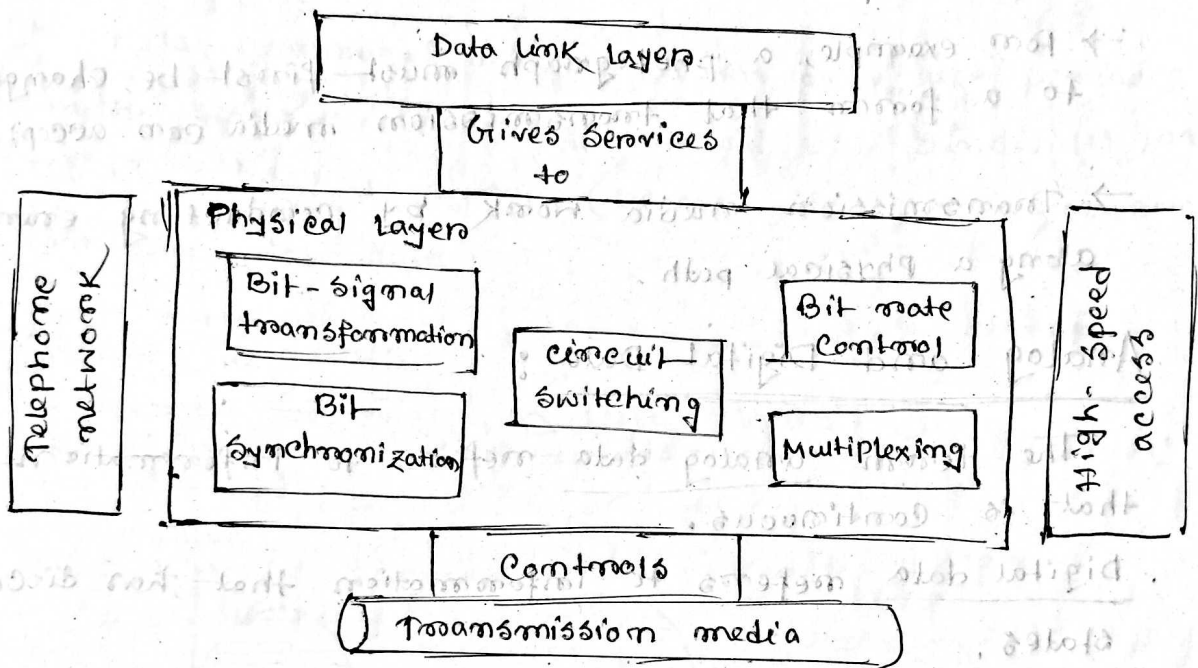
④ 100100

⑤ 011011

Various Devices in Computer Network

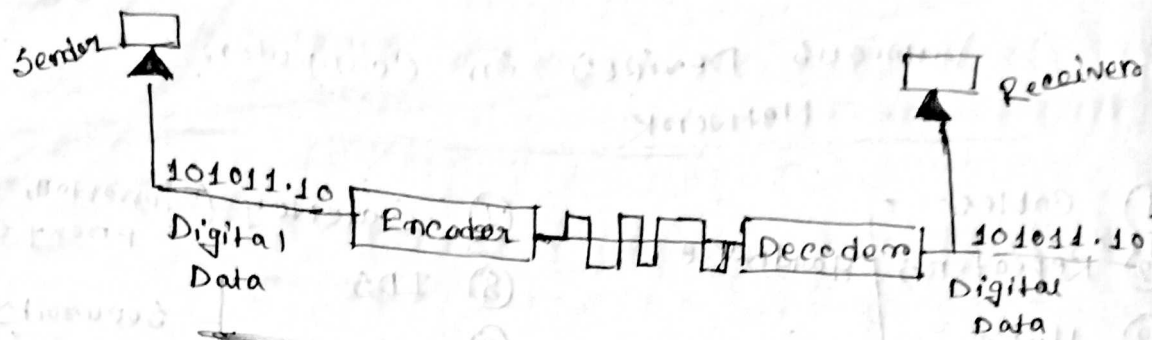
- | | | | |
|-------------|---------------------|--|----------|
| ① Cables | Hardware | ⑦ Gateway (firewall, router, proxy server) | Security |
| ② Repeaters | | ⑧ IDS | |
| ③ Hubs | | ⑨ Firewall | |
| ④ Bridges | Hardware & Software | ⑩ Modem (Modulator/Demodulator) | |
| ⑤ Switches | | | |
| ⑥ Routers | | | |

Position of the physical layer:



Duties of Physical Layer

- Bit-signal transformation
- Bit-rate Control
- Bit synchronization
- Multiplexing
- Circuit-switching



NOTE :

To be transmitted, data must be transformed to electromagnetic signals.

→ Generally, the data usable to a person or application are not in a form that can be transmitted over a network.

→ For example, a photograph must first be changed to a form that transmission media can accept.

→ Transmission media work by conducting energy along a physical path.

Analog and Digital Data :

The term analog data refers to information that is continuous.

Digital data refers to information that has discrete states.

Analog and Digital Signals :

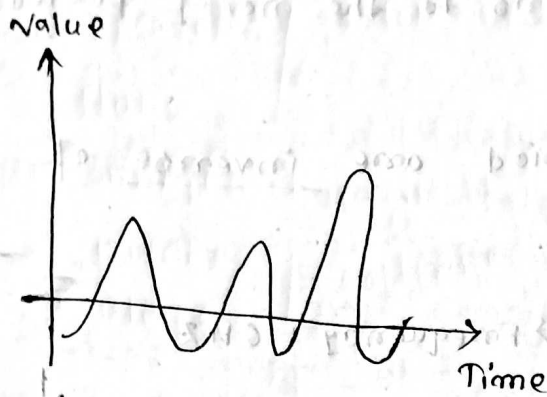
An analog signal has infinitely many levels of intensity over a period of time.

A digital signal, on the other hand, can have only a limited number of defined values.

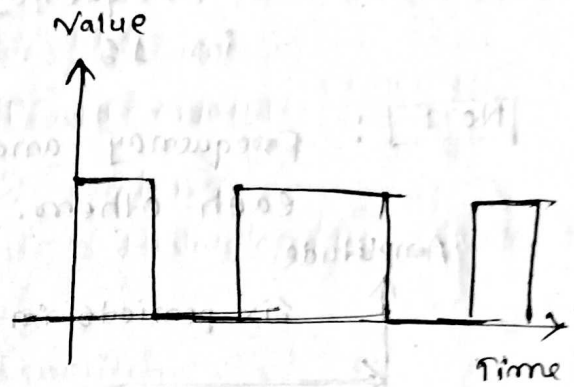
Periodic And Aperiodic signals :

A periodic signal completes a pattern within a measurable time frame, called a period, and repeats that pattern over subsequent identical periods.

The completion of one full pattern is called a cycle. A nonperiodic signal changes without exhibiting a pattern or cycle that repeats over time.



a. analog signal



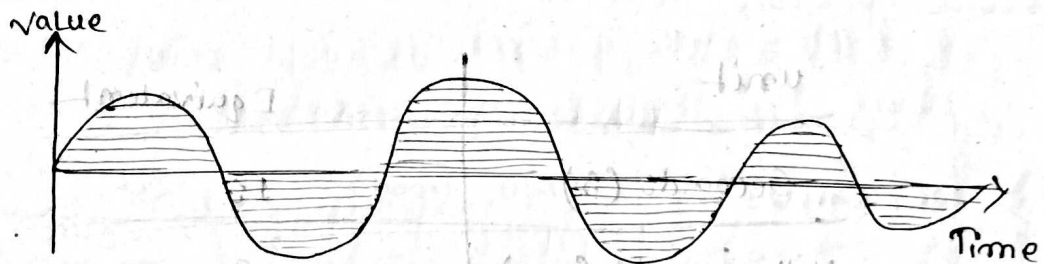
b. Digital signal

In data communication, we commonly use periodic analog signals (because they need less bandwidth) and aperiodic digital signals (because they can represent variation in data).

Analog signals

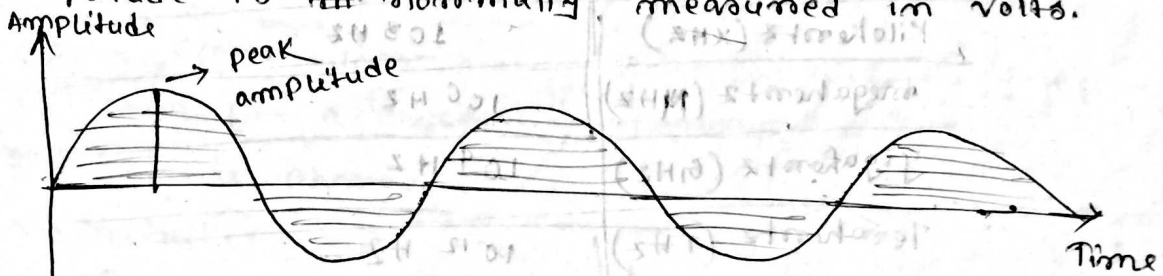
Sine wave

Sine wave:



peak amplitude:

The peak amplitude of a signal is the absolute value of its highest intensity, proportional to the energy it carries. For electric signals, peak amplitude is normally measured in volts.

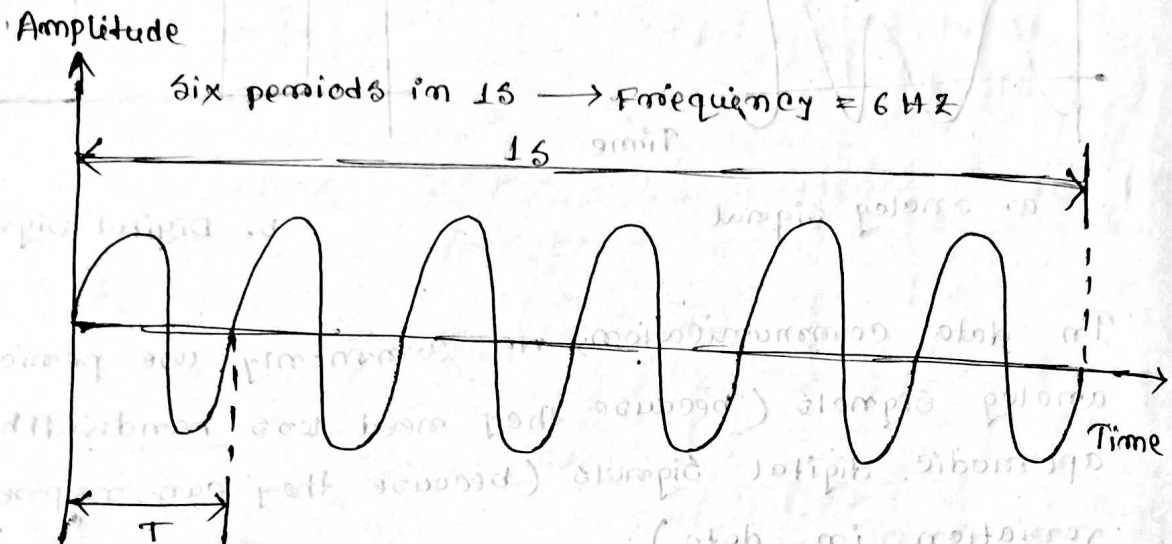


Period & Frequency

[Period] : period refers to the amount of time, in seconds, a signal needs to complete 1 cycle.

[Frequency] : Frequency refers to the no. of periods in 1s.

[NOTE] : Frequency and period are inverse of each other.



period = $\frac{1}{6}$ s

units of periods and frequencies :

unit	Equivalent
seconds (s)	1s
Milliseconds (ms)	10^{-3} s
Microseconds (μ s)	10^{-6} s
Nanoseconds (ns)	10^{-9} s
Picoseconds (ps)	10^{-12} s
Hertz (Hz)	1 Hz
Kilohertz (kHz)	10^3 Hz
Megahertz (MHz)	10^6 Hz
Gigahertz (GHz)	10^9 Hz
Terahertz (THz)	10^{12} Hz

Example

1. The power we use at home has a frequency of 60 Hz (50 Hz in Europe). The period of this time wave can be determined as follows:

$$\Rightarrow T = \frac{1}{f} = \frac{1}{60} = 0.0166\text{ s} \\ = 0.0166 \times 10^3 \text{ ms} = 16.6 \text{ ms}$$

This means that the period of the power for our lights at home is 0.0166 s or 16.6 ms. Our eyes are not sensitive enough to distinguish these rapid changes in amplitude.

2. Express a period of 100 ms in microseconds, and express the corresponding frequency in kHz.

$\Rightarrow$ We find the equivalent of 1 ms. We make the following substitutions:

$$100 \text{ ms} = 100 \times 10^{-3} \text{ s} = 100 \times 10^{-3} \times 10^6 \mu\text{s} = 10^5 \mu\text{s}$$

Now, we use the inverse relationship to find the frequency, changing Hertz to Kilohertz.

$$100 \text{ ms} = 100 \times 10^{-3} \text{ s} = 10^{-1} \text{ s}$$

$$f = \frac{1}{10^{-1} \text{ s}} = 10 \times 10^3 \text{ Hz} = 10^4 \text{ Hz}$$

NOTE

- (i) Frequency is the rate of change with respect to time. Change in a short span of time means high frequency.

Change over a long span of time means low frequency.

LF examples: power lines, electrical wiring and electrical appliances.

HF examples: X-rays, gamma rays.

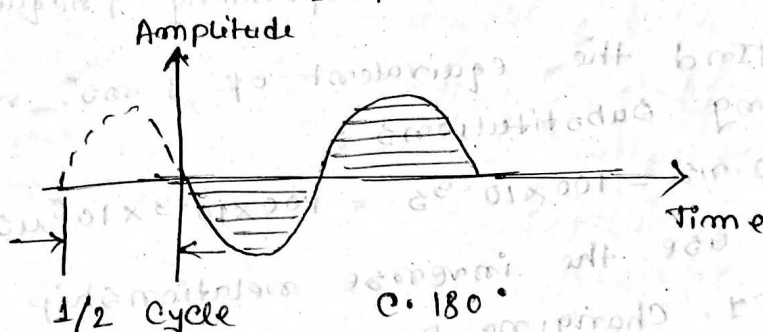
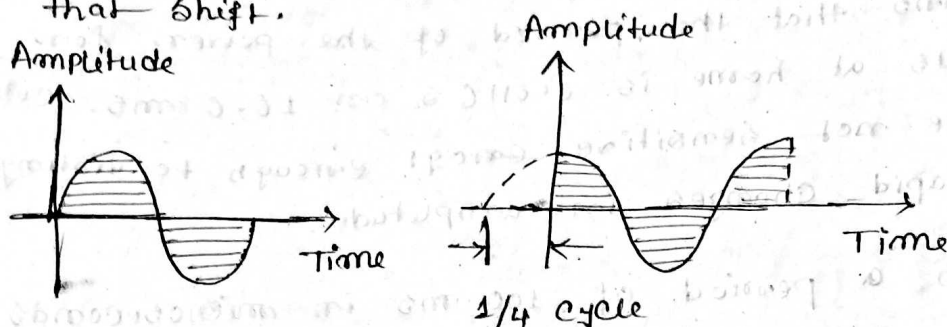
- (ii) If a signal does not change at all, its frequency is zero (ex. a constant voltage)

If a signal changes instantaneously, its frequency is infinite.

Phase

Phase describes the position of the waveform relative to time zero.

If we think of the wave as something that can be shifted backward or forward along the time axis, phase describes the amount of that shift.



Example: A sine wave is offset one-sixth of a cycle with respect to time zero. What is its phase in degrees and radians?

⇒ We know that one complete cycle is 360 degrees.

Therefore, $1/6$ cycle is,

$$1/6 \times 360 = 60 \text{ degrees}$$

Now,

$$2\pi \text{ rad} = 360^\circ$$

$$= \frac{360^\circ}{2\pi} \times \frac{180^\circ \times \pi}{22}$$

$$= \frac{60 \times 2\pi}{360} \text{ rad}$$

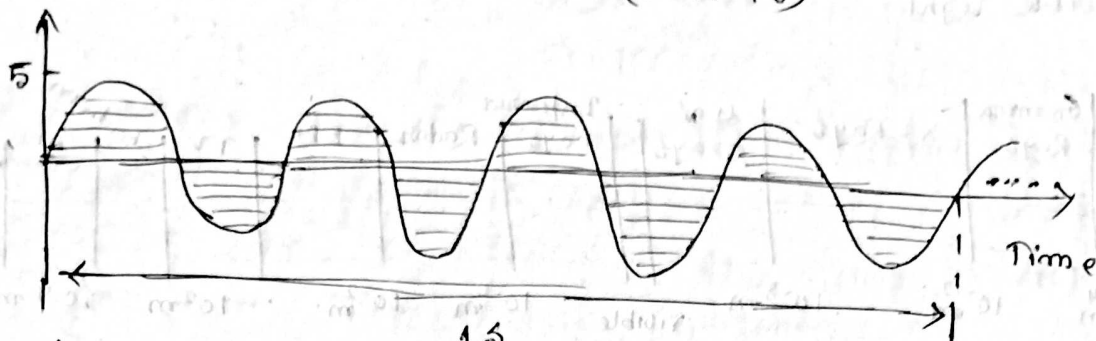
$$= 1.046 \text{ rad}$$

Sine wave examples :

$$A = 5 \quad f = 4 \quad \phi = 0$$

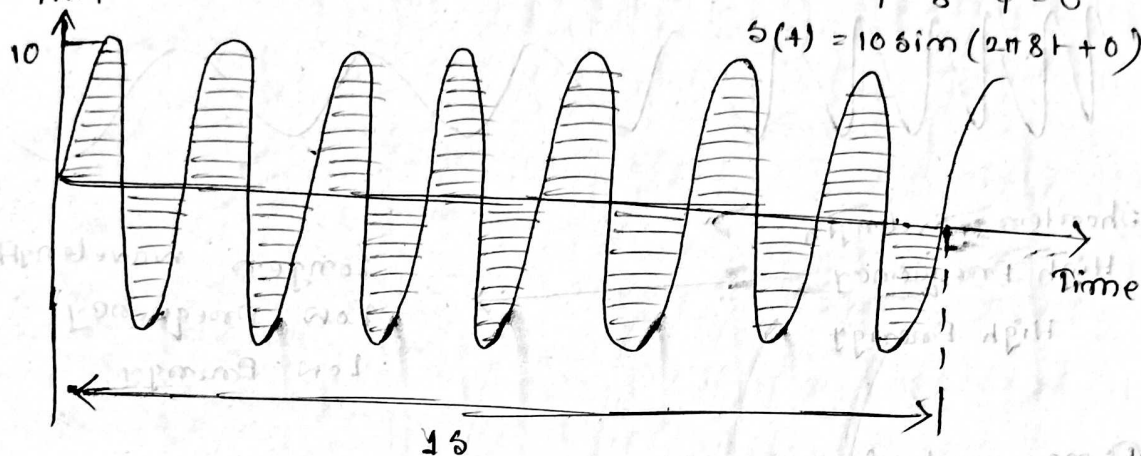
$$s(t) = 5 \sin(2\pi 4t + 0)$$

Amplitude



Continued

Amplitude

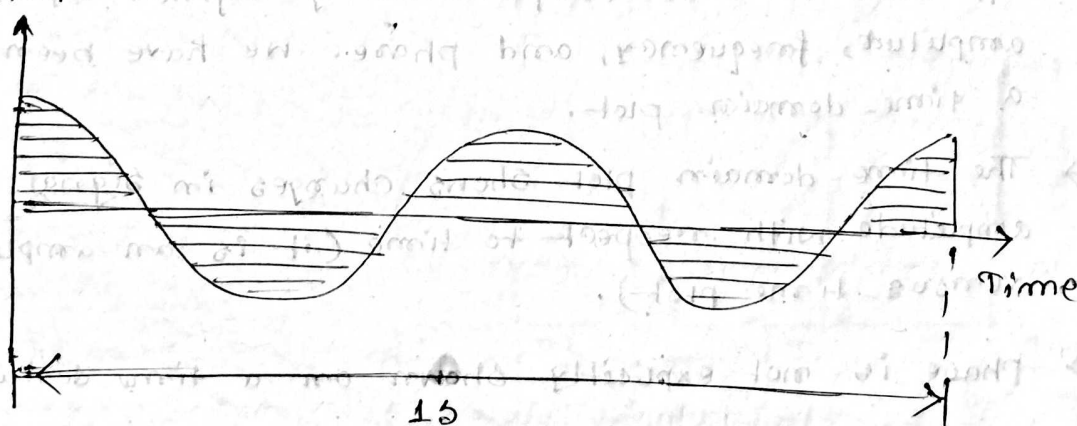


$$A = 10 \quad f = 8 \quad \phi = 0$$

$$s(t) = 10 \sin(2\pi 8t + 0)$$

Continued

Amplitude

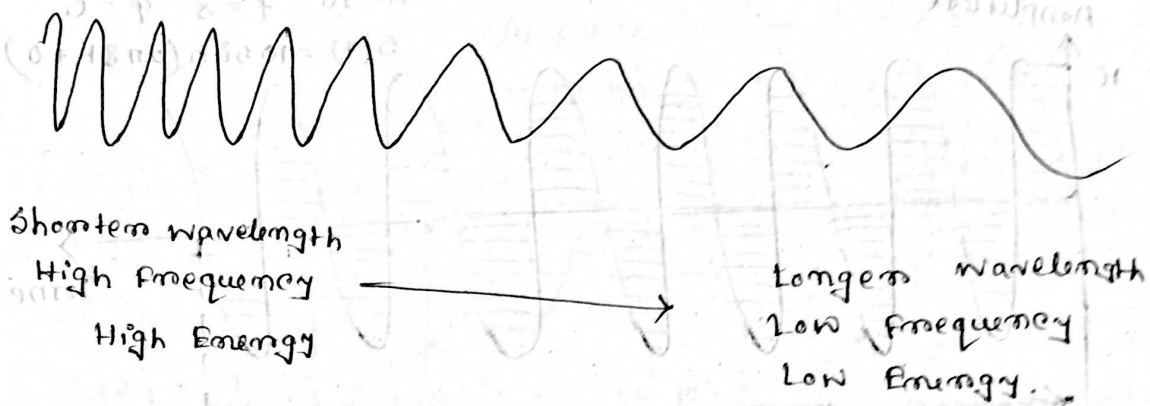
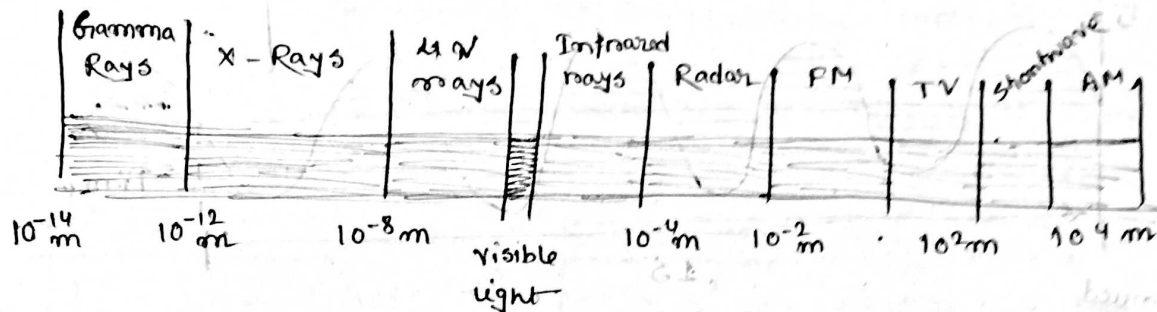


$$A = 5 \quad f = 2 \quad \phi = \pi/2$$

$$s(t) = 5 \sin(2\pi 2t + \pi/2)$$

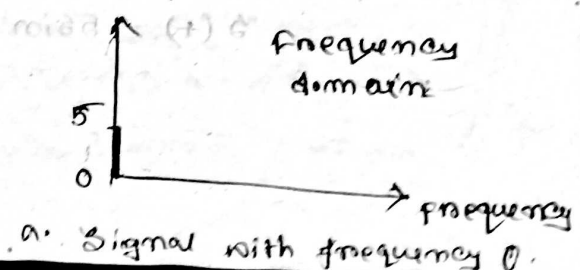
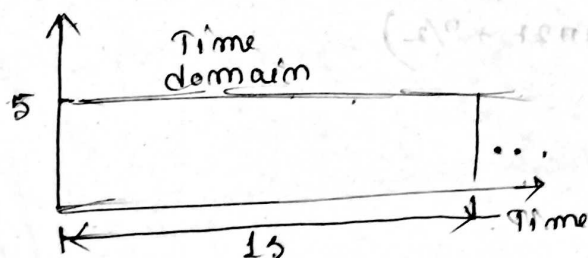
Wavelength : Wavelength is the distance between identical points in the adjacent cycle of a waveform. Signal propagated in space or along a wire.

Wavelength of visible light = 380nm - 750nm



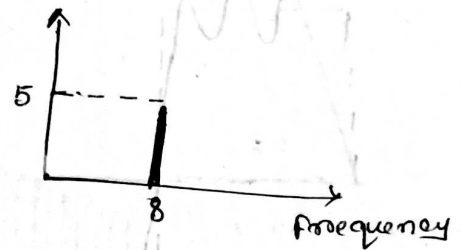
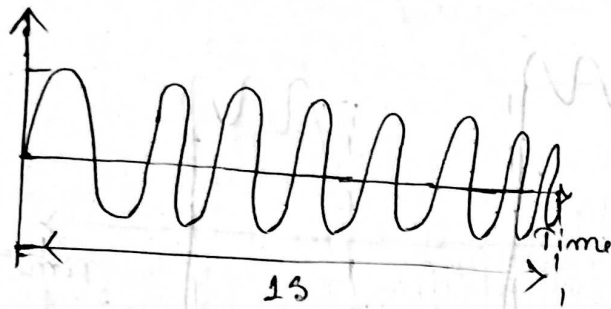
Time and frequency domains

- A sine wave is comprehensively defined by its amplitude, frequency, and phase. We have been showing a time-domain plot.
- The time-domain plot shows changes in signal amplitude with respect to time (it is an amplitude-versus-time plot).
- Phase is not explicitly shown on a time-domain plot.
- A frequency-domain plot is ~~also~~ concerned with only the peak value and the frequency. Changes of amplitude during one period are not shown.



a. Signal with frequency 0.

→ A complete sine wave in the time domain can be represented by one single spike in the frequency domain.



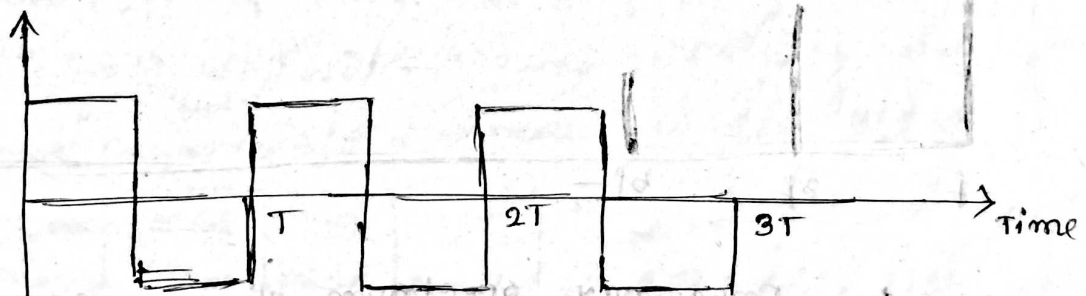
b. A signal with frequency 8

NOTE : (i) A single-frequency sine wave is not useful in data communications, we need to change one or more of its characteristics to make it useful.

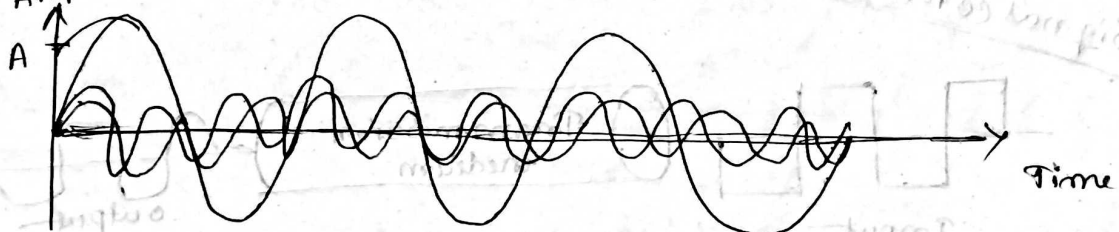
(ii) When we change one or more characteristics of a single-frequency signal, it becomes a composite signal made of many frequencies.

(iii) According to Fourier analysis, any composite signal can be represented as a combination of simple sine waves with different frequencies, phases and amplitudes.

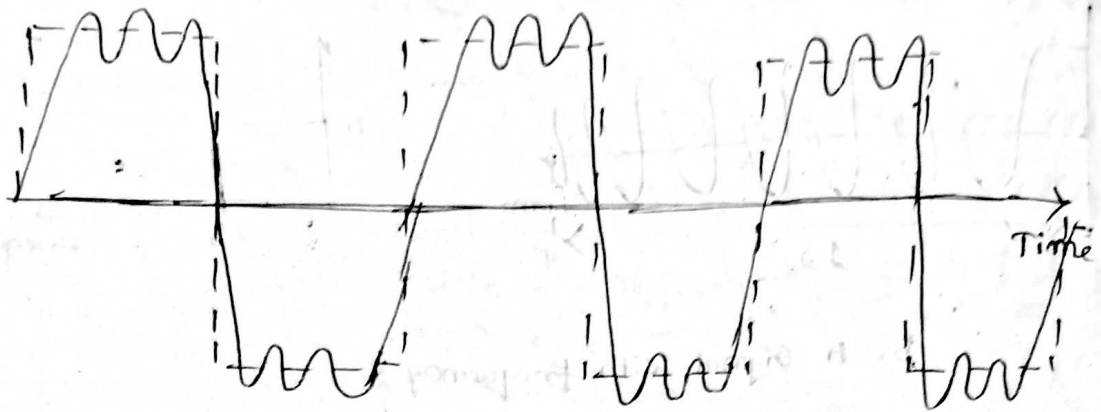
Square wave



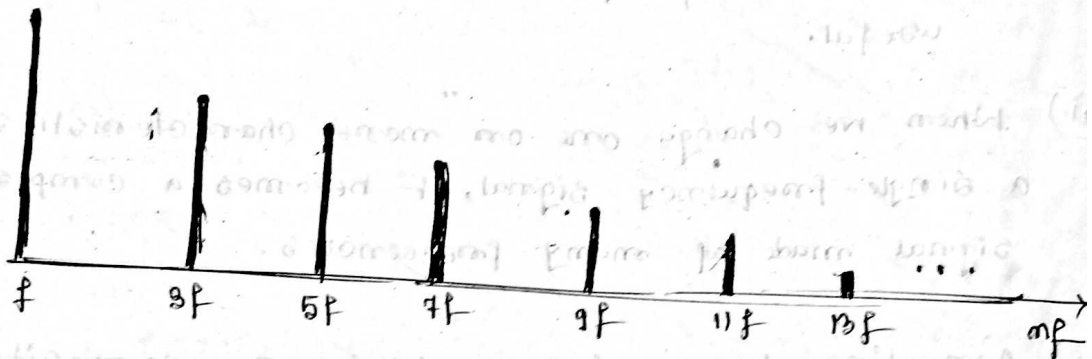
Three Harmonics
Amplitude



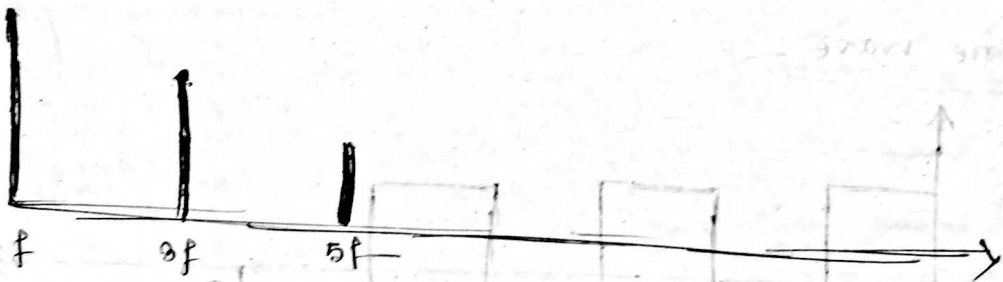
1] Adding first three harmonics :



Frequency Spectrum Comparison :

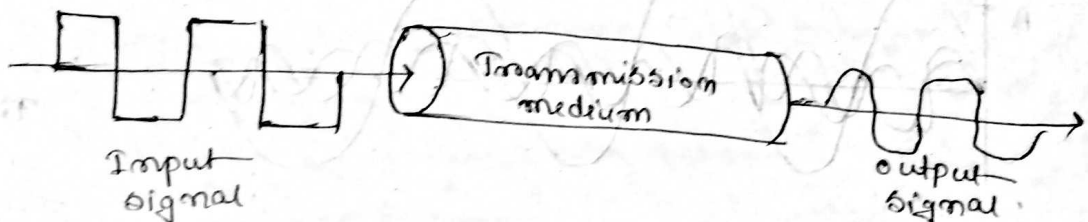


a. Frequency Spectrum of a Square Wave

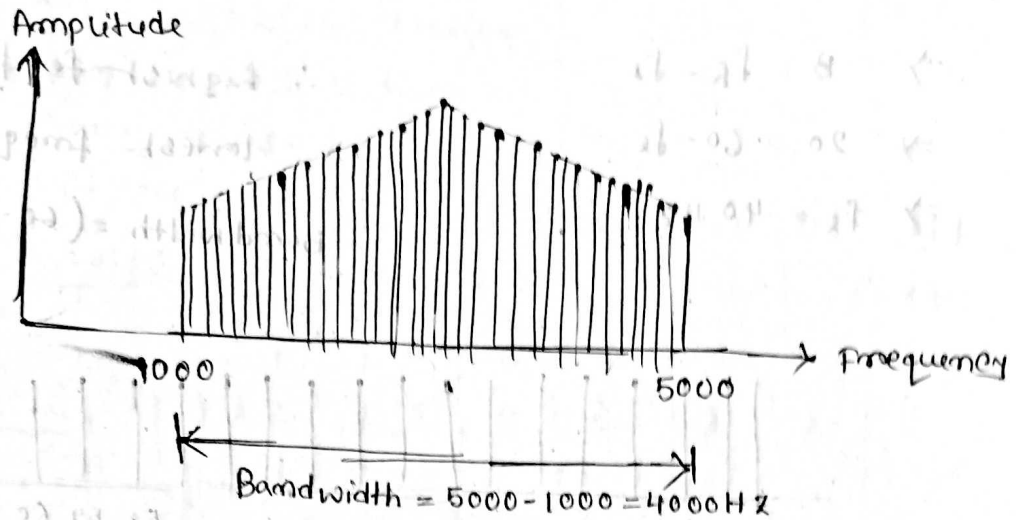


b. Frequency spectrum of an approximation with only three harmonics

Signal corruption



Bandwidth : The bandwidth is a property of a medium. It is the difference between the highest and the lowest frequencies that the medium can satisfactorily pass.



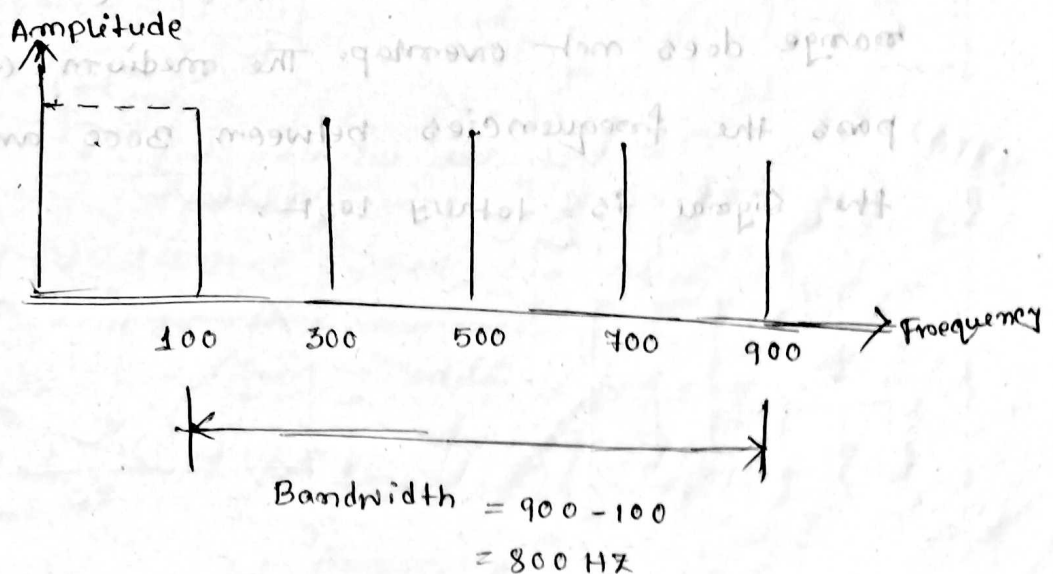
Example :

1. If a periodic signal is decomposed into five sine waves with frequencies of 100, 300, 500, 700 and 900 Hz, what is the bandwidth? Draw the spectrum, assuming all components have a maximum amplitude of 10 V.

$$\Rightarrow B = f_h - f_l = 900 - 100 = 800 \text{ Hz}$$

$$\therefore \text{Bandwidth} = 800 \text{ Hz}$$

The spectrum has only five spikes at 100, 300, 500, 700 and 900 Hz.



2. A signal has a bandwidth of 20 Hz. The highest frequency is 60 Hz. What is the lowest frequency? Draw the spectrum if the signal contains all integral frequencies of the same amplitude.

$$\Rightarrow B = f_h - f_l$$

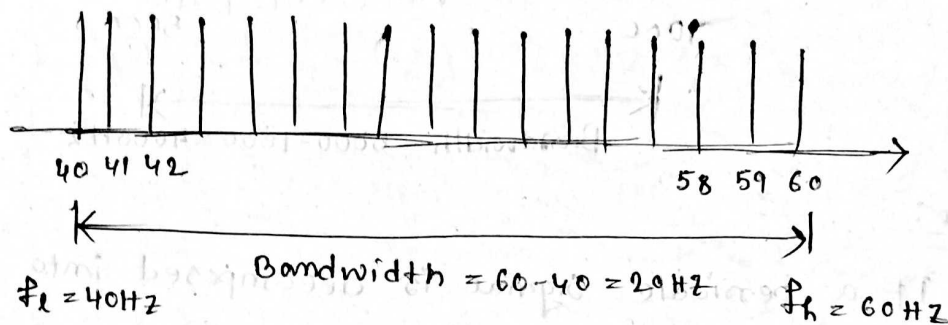
$$\therefore \text{highest freq} = 60 \text{ Hz}$$

$$\Rightarrow 20 = 60 - f_l$$

$$\therefore \text{lowest freq} = 40 \text{ Hz}$$

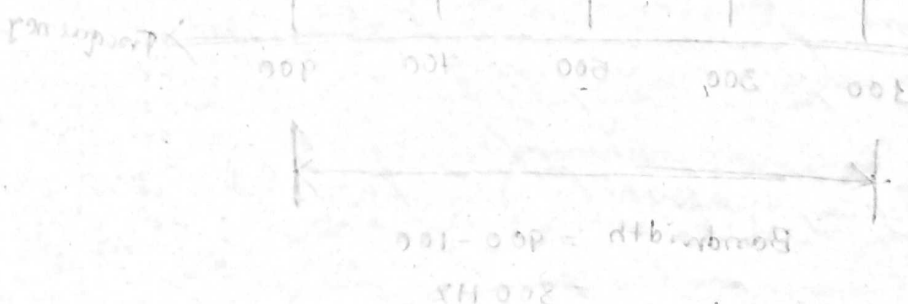
$$\Rightarrow f_l = 40 \text{ Hz}$$

$$\therefore \text{Bandwidth} = (60 - 40) = 20 \text{ Hz}$$



3. A signal has a spectrum with frequencies between 1000 and 2000 Hz (bandwidth of 1000 Hz). A medium can pass frequencies from 3000 to 4000 Hz (a bandwidth of 1000 Hz). Can this signal faithfully pass through this medium?

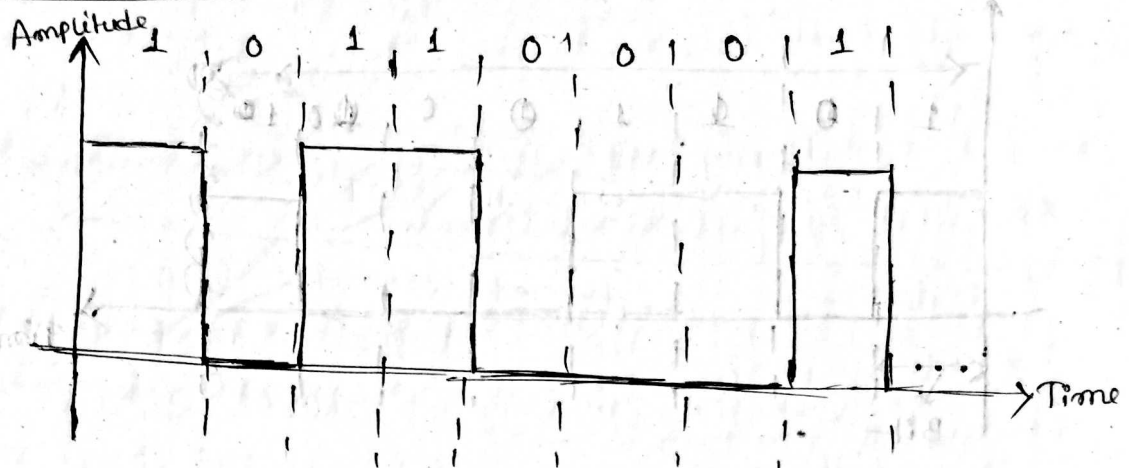
$\Rightarrow$ The answer is definitely no. Although the signal can have the same bandwidth (1000 Hz), the range does not overlap. The medium can only pass the frequencies between 3000 and 4000 Hz. The signal is totally lost.



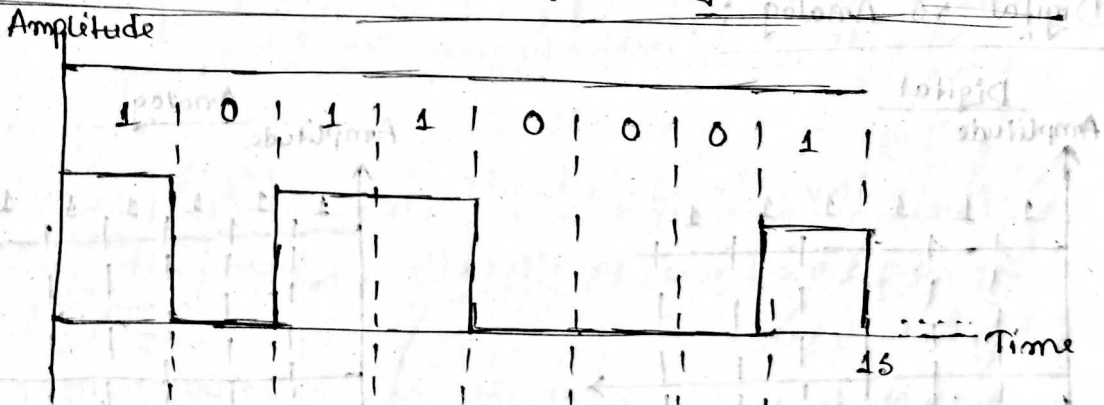
Digital Signal

- Bit Interval and Bit Rate
- As a Composite Analog Signal
- Through wide-Bandwidth Medium
- Through Band-Limited Medium
- Versus Analog Bandwidth
- Higher Bit Rate

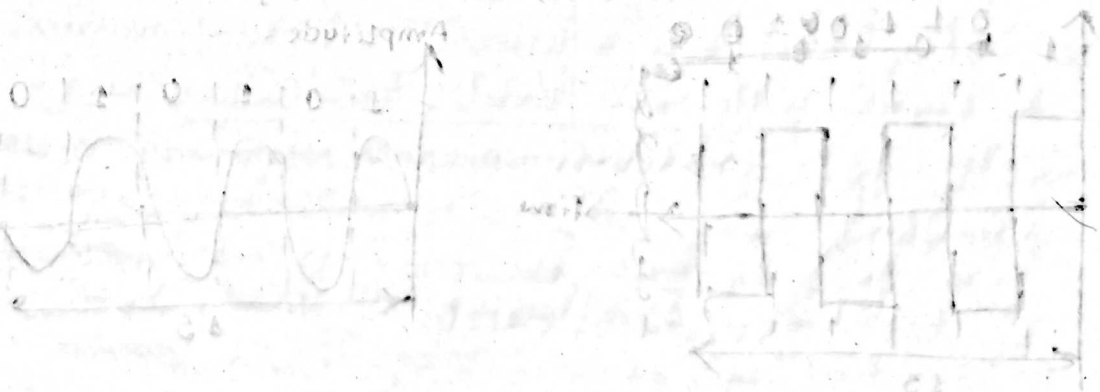
A Digital Signal:



Two digital signals: one with two signal levels and the other with four signal levels



check
PDP



Example:

A digital signal has a bit rate of 2000 bps. What is the duration of each bit (bit interval)?

→ The bit interval is the inverse of the bit rate.

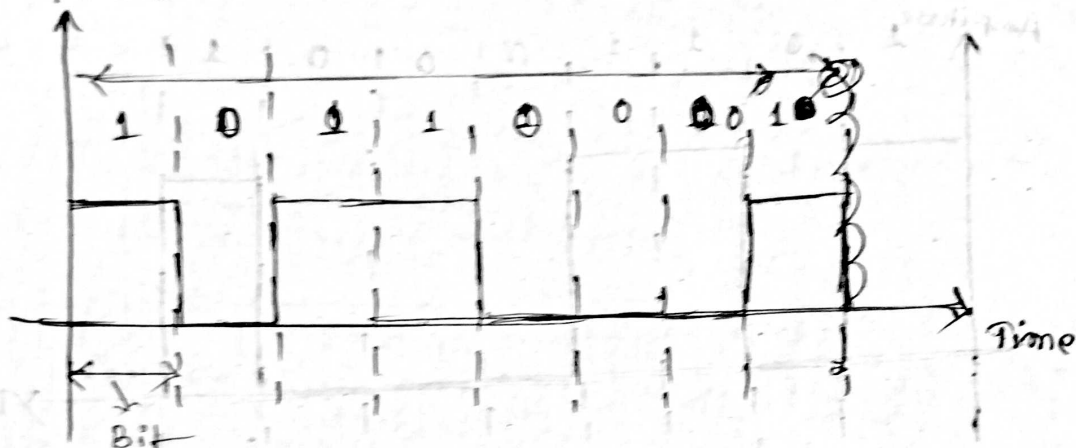
$$\text{Bit interval} = \frac{1}{2000 \text{ s}} = 0.000500 \text{ s}$$

$$= 0.000500 \times 10^6 \mu\text{s}$$

$$= 500 \mu\text{s}$$

Bit rate and bit interval:

Amplitude



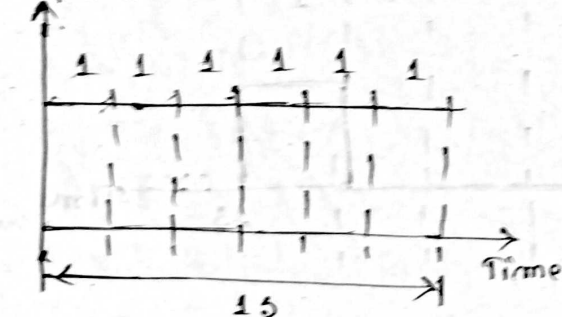
Interval

$$1 \text{ s} = 8 \text{ bit intervals}$$

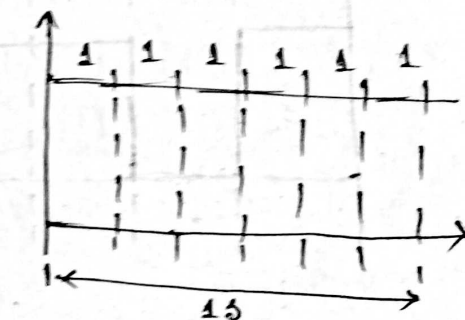
$$\text{Bit rate} = 8 \text{ bps}$$

Digital vs Analog:

Digital
Amplitude

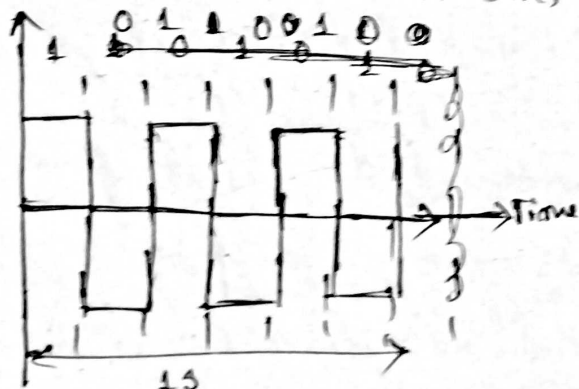


Analog
Amplitude

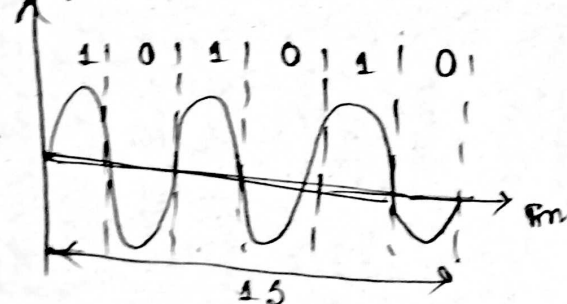


101010

a. Best case, bit rate = 6, $f = 0$



Amplitude



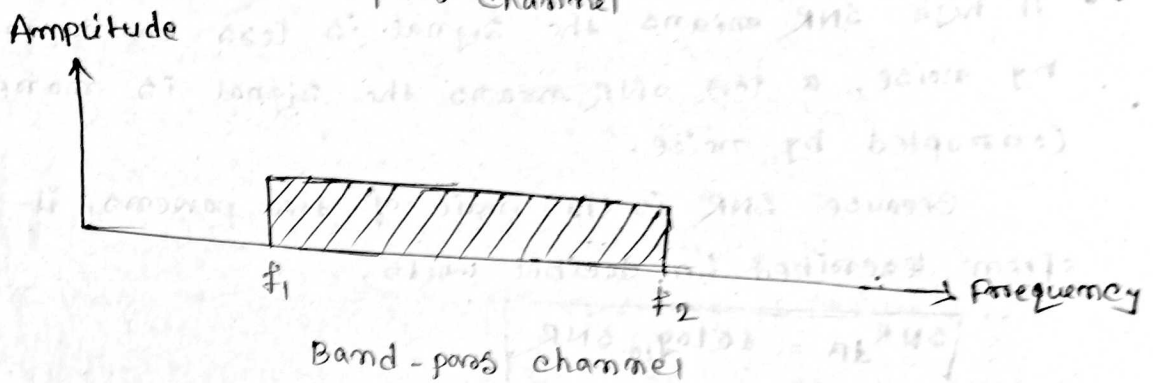
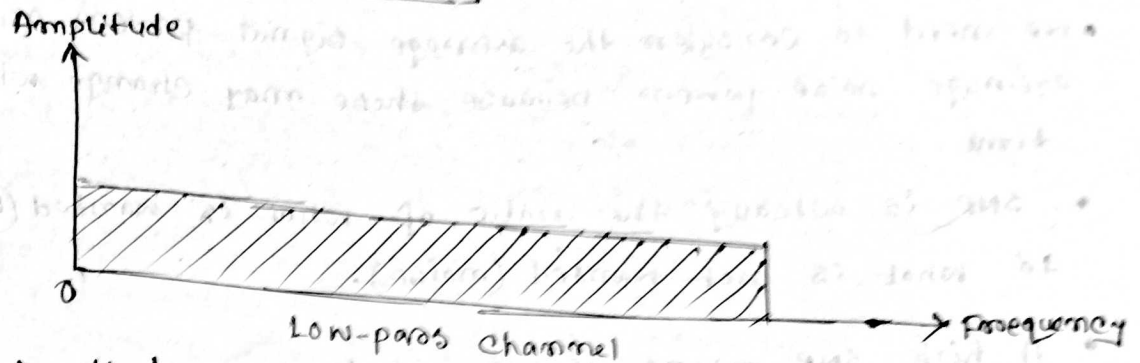
b. Worst case, bit rate = 6, $f = 3$

[NOTE]: The bit rate and the bandwidth are proportional to each other.

Analog vs Digital

- Low-pass Versus Band-pass
- Digital Transmission
- Analog Transmission

Low-pass and Band-pass



[NOTE]: (i) The analog bandwidth of a medium is expressed in Hertz, the digital bandwidth, in bits per second.

(ii) Digital transmission needs a low-pass channel.

(iii) Analog transmission can use a band-pass channel.

Data Rate Limit

Noiseless channel: Nyquist Bit Rate

$$\text{BitRate} = 2 \times \text{bandwidth} \times \log_2 L$$

Noisy channel: Shannon Capacity

$$\text{Capacity} = \text{Bandwidth} \times \log_2 (1 + \text{SNR})$$

In practice, we need to use both methods to find the limits and signal levels.

SNR (Signal-to-Noise Ratio)

As we will see later, to find the theoretical bit rate limit, we need to know the ratio of the signal power to the noise power. The signal-to-noise ratio is defined as —

$$\text{SNR} = \frac{\text{avg. signal power}}{\text{avg. noise power}}$$

- we need to consider the average signal power and the average noise power because these may change with time.
- SNR is actually the ratio of what is wanted (signal) to what is not wanted (noise).
- A high SNR means the signal is less corrupted by noise, a low SNR means the signal is more corrupted by noise.

Because SNR is the ratio of two powers, it is often described in decibel units,

$$\text{SNR}_{\text{dB}} = 10 \log_{10} \text{SNR}$$

Example:

1. Consider a noiseless channel with a bandwidth of 3000 Hz, transmitting a signal with two signal levels. The maximum bit rate can be calculated as —

$$\text{Bit Rate} = 2 \times 3000 \times \log_2 2 = 6000 \text{ bps.}$$

2. Consider the same noiseless channel, transmitting a signal with four levels (for each level, we send two bits). The maximum bit rate can be calculated as —

$$\text{Bit Rate} = 2 \times 3000 \times \log_2 4 = 12,000 \text{ bps.}$$

3. Consider an extremely noisy channel in which the value of the signal-to-noise ratio is almost zero. In other words, the noise is so strong that the signal is faint. For this channel the capacity is calculated as —

$$C = B \log_2 (1 + \text{SNR}) = B \log_2 (1 + 0) \\ = B \log_2 (1) = B \times 0 = 0$$

4. We can calculate the theoretical highest bit rate of a regular telephone line. A telephone line normally has a bandwidth of 3000 Hz (300 Hz to 3300 Hz). The signal-to-noise ratio is usually 3162. For this channel the capacity is calculated as —

$$C = B \log_2 (1 + \text{SNR}) = 3000 \log_2 (1 + 3162) \\ = 3000 \log_2 (3163)$$

$$\therefore C = 3000 \times 11.62 \\ = 34,860 \text{ bps.}$$

5. We have a channel with a 1 MHz bandwidth. The SNR for this channel is 63, what is the appropriate bit rate and signal level?

⇒ First, we use the Shannon formula to find our upper limit.

$$C = B \log_2 (1 + \text{SNR}) = 10^6 \log_2 (1 + 63) \\ = 10^6 \log_2 (64) \\ = 6 \text{ Mbps.}$$

Then we use the Nyquist formula to find the number of signal levels.

$$4 \text{ Mbps} = 2 \times 1 \text{ MHz} \times \log_2 L$$

$$\Rightarrow L = 4$$

Transmission Impairment

Transmission impairment occurs when the received signal is different from the transmitted signal. As we know, a signal can be transmitted as Analog signal or it can be transmitted as a digital signal.

In analog signals due to transmission impairment the resulting received signal gets

different amplitude or the shape. In the case of

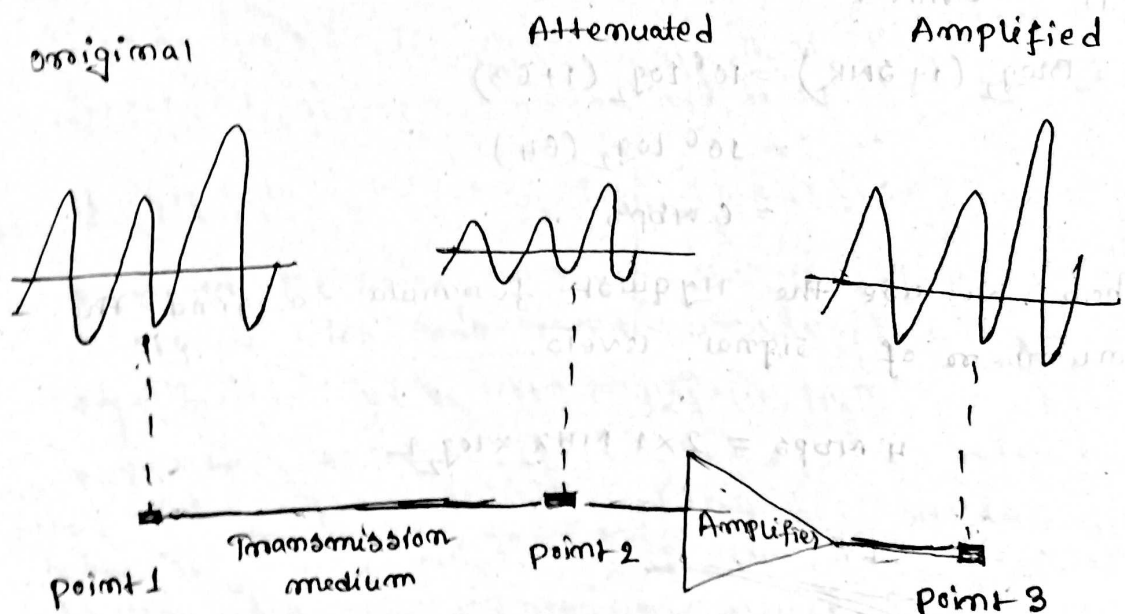
digitally transmitted signals at the receiver side we get changes in bits (0's or 1's).

There are various causes/types of transmission impairments ————

- (i) Noise
- (ii) Distortion
- (iii) Attenuation

Attenuation :

Attenuation is the loss of signal strength in networking cables or connections. This typically is measured in decibels (dB) or voltage and can occur due to a variety of factors. It may cause signals to become distorted or indistinguishable.



Ex: ① Imagine a signal travels through a transmission medium and its power is reduced to half. This means that $P_2 = \frac{1}{2} P_1$. In this case, the attenuation (loss of power) can be calculated as —

$$10 \log_{10} (P_2/P_1)$$

$$\rightarrow 10 \log_{10} (0.5 P_1 / P_1) = 10 \log_{10} (0.5)$$

$$= 10 (-0.3) = -3 \text{ dB}$$

② Imagine a signal travels through an amplifier and its power is increased ten times. This means that $P_2 = 10 \times P_1$. In this case, the amplification (gain of power) can be calculated as —

$$10 \log_{10} (P_2/P_1)$$

$$= 10 \log_{10} (10 P_1 / P_1)$$

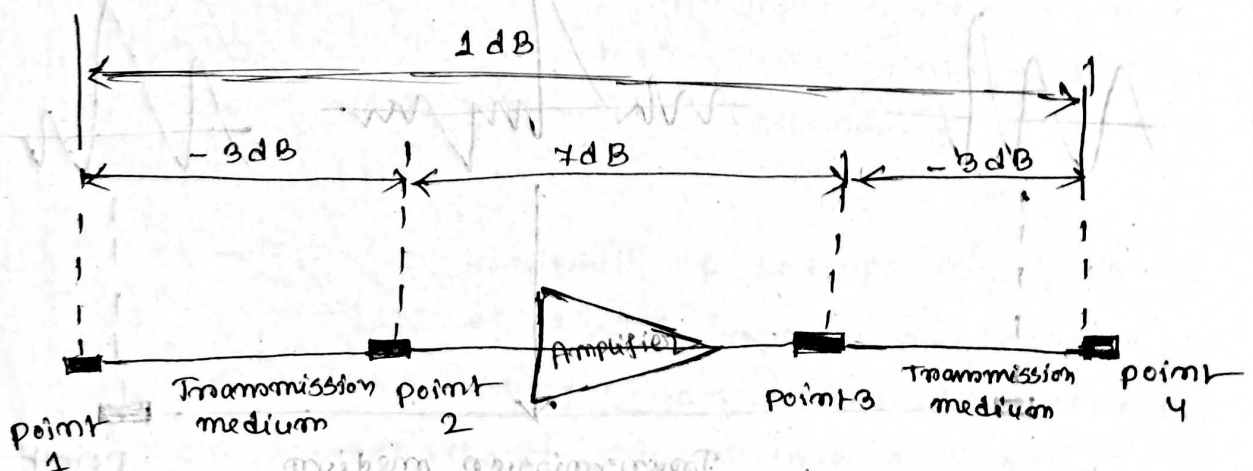
$$= 10 \log_{10} (10)$$

$$= 10(1)$$

$$= 10 \text{ dB}$$

③ SEE PPT Physical-Layer.pdf page 68

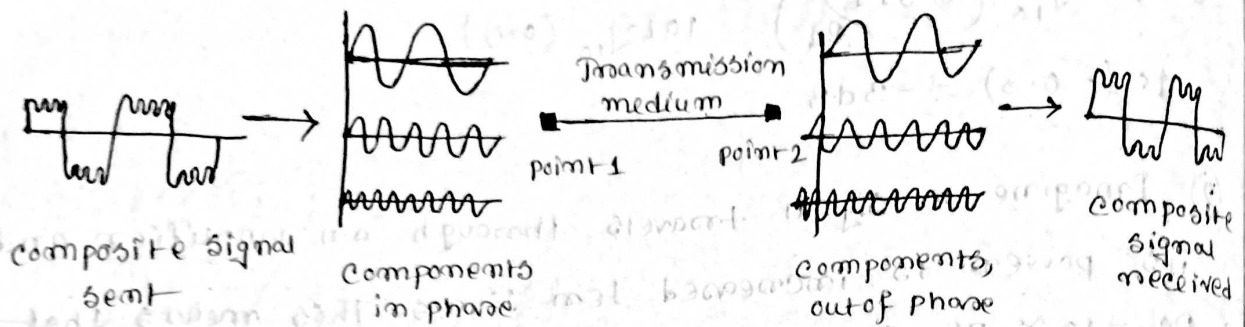
④



$$\text{dB} = -3 + 7 - 3 = +1$$

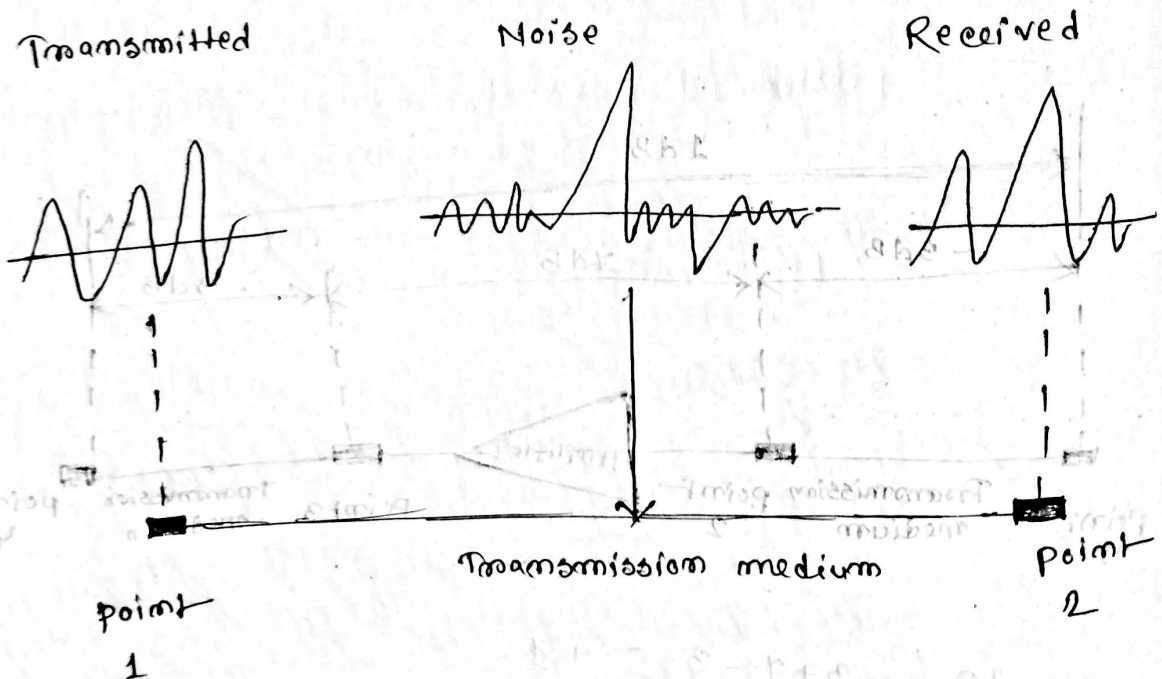
Distortion :

Distortion, in acoustics and electronics, any change in a signal that alters the basic waveform or the relationship between various frequency components. It is usually a degradation of the waveform is not usually considered to be distortion.



Noise :

Noise is the major factor for the transmission distortion as any unwanted signal gets added to the transmitted signal by which the resulting transmitted signal gets modified and at the receiver side it is difficult to remove the unwanted noise signal. These noises are various kinds like shot noise, impulse noise, thermal noise etc.

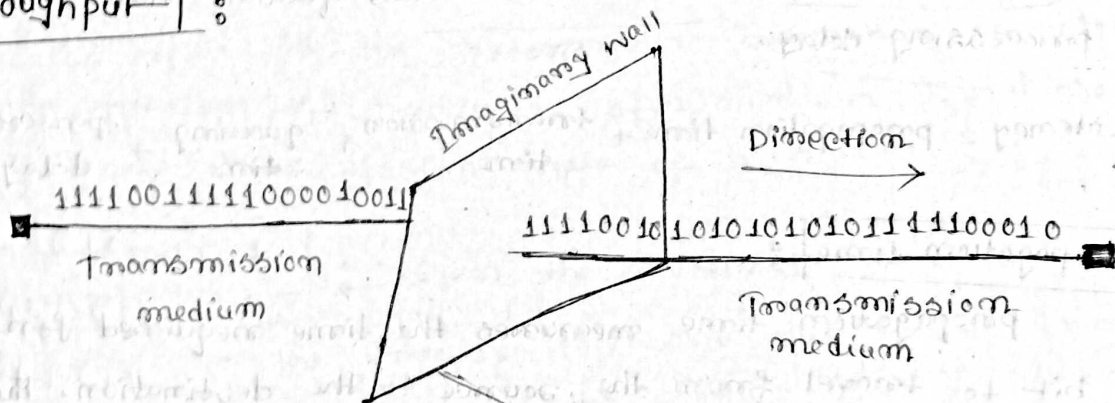


Bandwidth: One characteristic that measures network performance is bandwidth. However, the term can be used in two different contexts with two different measuring values: bandwidth is Hz and bandwidth in bits per second.

In networking, we use the term bandwidth in two contexts

- Bandwidth in Hz, refers to the range of frequencies in a composite signal or the range of frequencies that a channel can pass.
- Bandwidth in bits per second, refers to the speed of bit transmission in a channel or link.

Throughput:



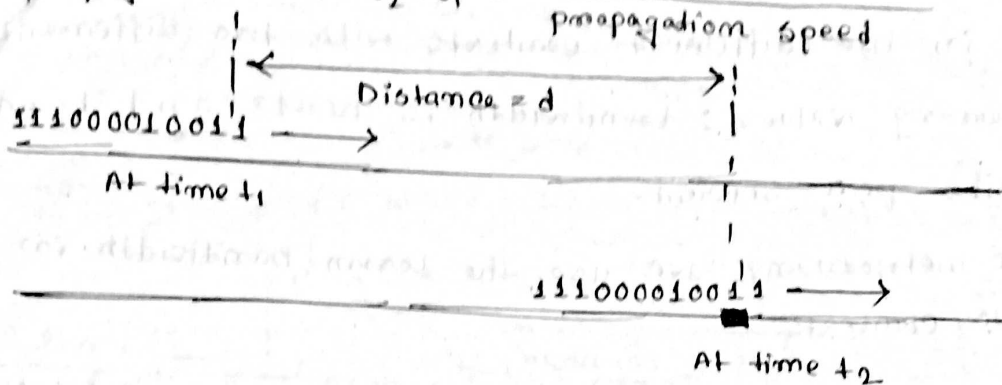
Throughput is the number of bits passing through this wall in a second.

Ex: A network with bandwidth of 10 Mbps can pass only an average of 12,000 frames per minute with each frame carrying an average of 10,000 bits. What is the throughput of this network?

$$\Rightarrow \text{Throughput} = \frac{12,000 \times 10,000}{60} \\ = 2 \text{ Mbps.}$$

Propagation time:

$$\text{propagation time} = t_2 - t_1 = \frac{\text{Distance } d}{\text{propagation speed}}$$



Latency (Delay):

The latency or delay defines how long it takes for an entire message to completely arrive at the destination from the time the first bit is sent out from the source. We can say that latency is made of four components — propagation time, transmission time, queuing time and processing delay.

$$\text{Latency} = \text{propagation time} + \text{transmission time} + \text{queuing time} + \text{processing delay}.$$

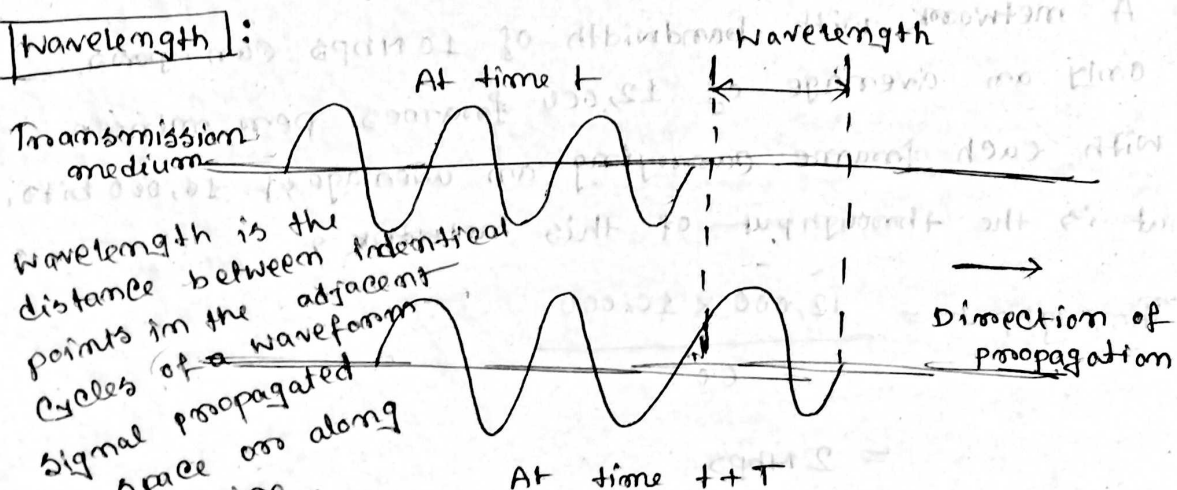
propagation time:

propagation time measures the time required for a bit to travel from the source to the destination. The propagation time is calculated by dividing the distance by the propagation speed.

$$\text{propagation time} = \frac{\text{Distance}}{\text{propagation speed}}.$$

Wavelength:

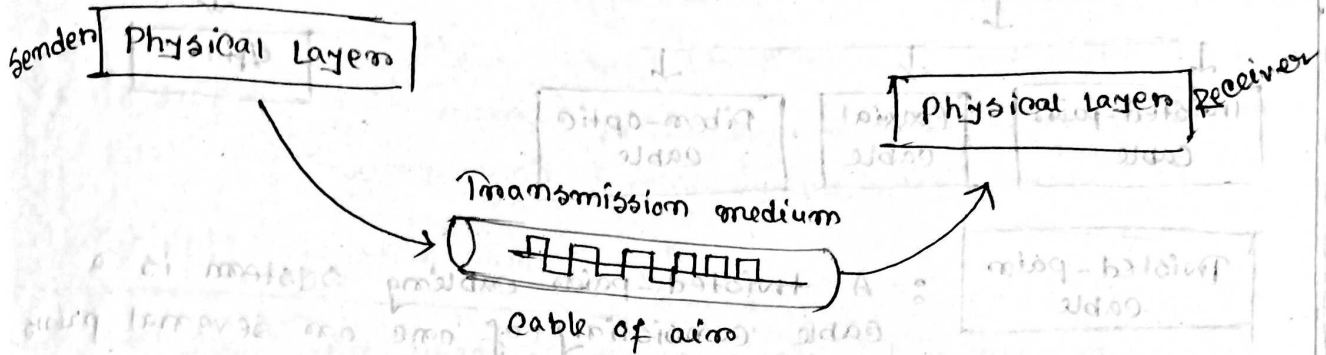
Wavelength is the distance between identical points in the adjacent cycles of a waveform signal propagated in space or along a wire.



Transmission medium and Physical layer

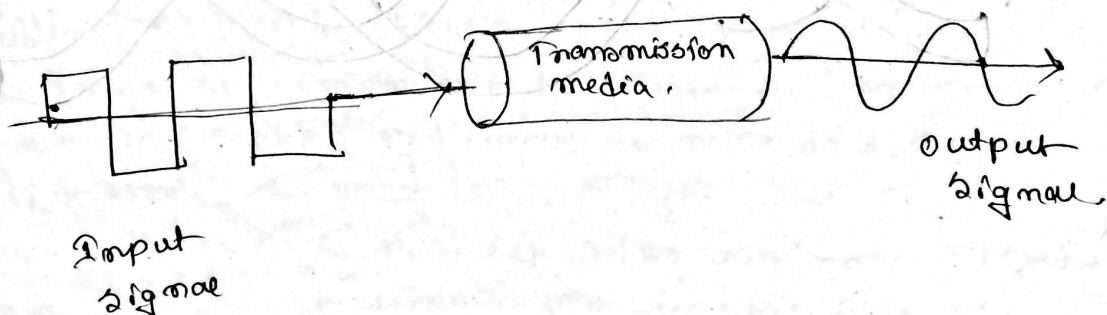
Transmission media

Transmission media is a communication channel that carries the information from the sender to the receiver. Data is transmitted through the electromagnetic signals.

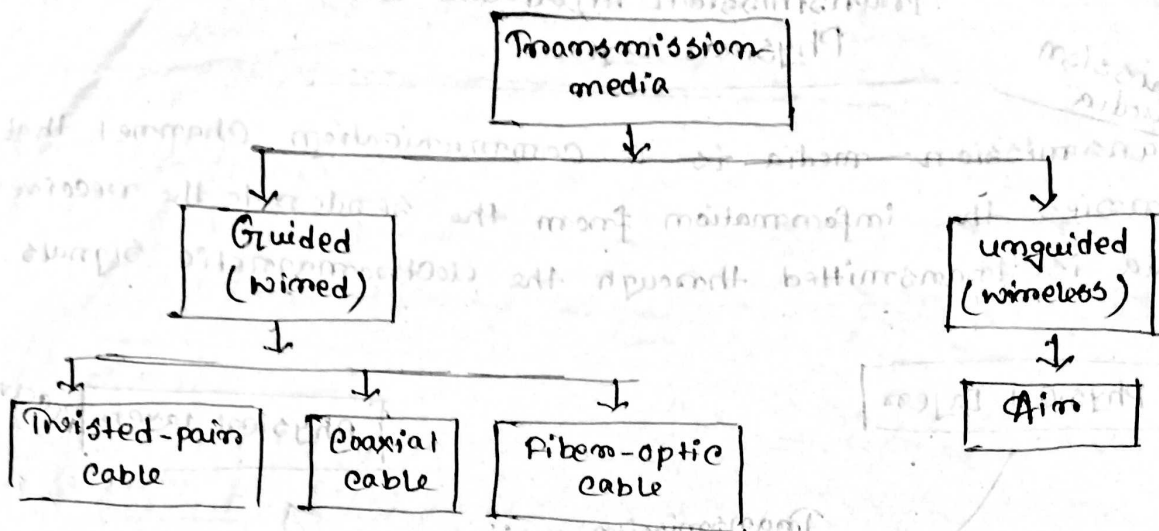


Factors need to be considered for designing the transmission media

- (i) **Bandwidth**: All the factors are remaining constant, the greater the bandwidth of a medium, the higher, the data transmission rate of a signal.
- (ii) **Transmission impairment**: When the received signal is not identical to the transmitted one due to the transmission impairment. The quality of the signal will get destroyed due to the transmission impairment.
- (iii) **Interference**: An interference is defined as the process of disrupting a signal when it travels over a communication medium on the addition of some unwanted signal.



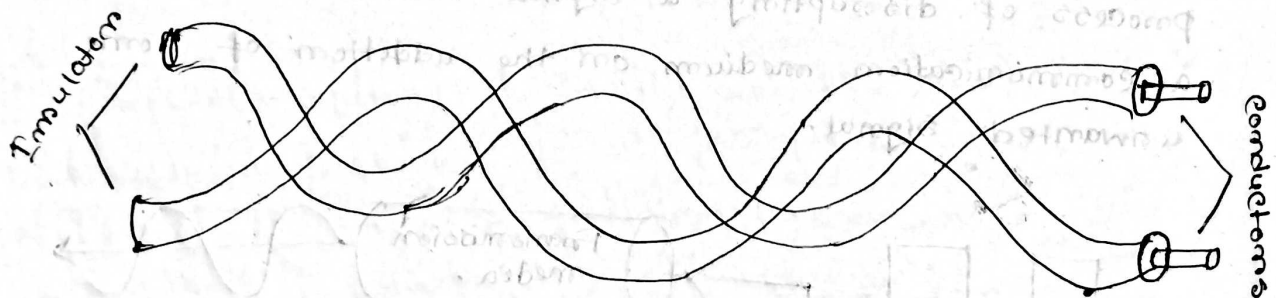
Classes of transmission media :-



Twisted-pair cable

: A twisted-pair cabling system is a cable consisting of one or several pairs of copper wires. These wires are twisted together and around each other and are insulated with a dielectric polymeric compound. Twisting helps minimize electromagnetic radiation and resists external interference. It also helps limit interference with adjacent twisted-pair cables.

Twisted-pair cables were among the earliest guided transmission media along with coaxial cables. Today, twisted-pair cables are used in many applications in networking and communication, for example, in telephone lines, Digital subscriber line and local area networks.



Q: Why is there twisting in twisted-pair cables?

→ To prevent noise, signal interference and crosstalk during transmissions, twisting is introduced into conducting cables. Twisting reduces the strength of noise signals and cancels out external waves by moving a part of the noise in the direction of the signal and another part in the opposite direction. Since the receiver can calculate the voltage difference of two wires to retrieve data more efficiently, twisting provides better noise immunity and lower signal attenuation.

Unshielded twisted pair (UTP)

As the name suggests, a UTP cable has no internal shielding. The UTP cable is the most common twisted-pair cable used in Ethernet networks. It is also used for traditional telephone (UTP-cat4) systems.

UTP cables are popular all over the world due to their low cost, ease of installation and flexibility. They also support high data transfer speeds of up to 1 Gbps per second (Gbps) for transmission distances of up to 100 meters (m). These cables are also easy to expand and troubleshoot.

One crucial drawback of UTP is that it is vulnerable to signal interference, which can affect output quality. Moreover, it is not suitable for long-distance transmission. To support distances longer than 100 m, repeaters are required.

Shielded twisted pair (STP)

STP cables include a shielding, usually made of aluminum foil, around the conductors to isolate the conductors and improve the cables' resistance to noise and signal interference.

One drawback is that STP cables are more difficult to connect at termination points. Also, STP

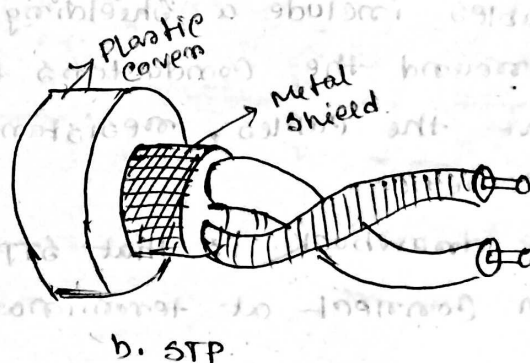
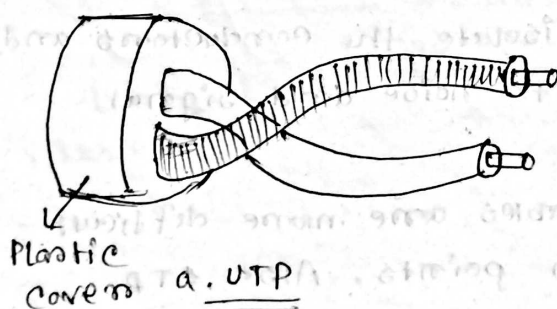
Cables are more expensive than UTP cables. Nonetheless, they prevent signal interference better than UTP cables, so their additional cost can be worth it, depending on the application.

Shielded cables support data transfer rates of 10 to 100 megabits per second (Mbps). They are best suited for transmission distances of up to 100m.

After that see ppt. many things are there.

Difference between UTP & STP cables

Parameters	UTP cable	STP Cable
Structure	These cables have twisted pairs of copper wires.	These cables have twisted pair of copper wires enclosed in a metal foil.
Form	unshielded twisted pair cable.	Shielded twisted pair cable.
Cost	Relatively cheaper than STP cables.	More expensive than UTP cables.
Electrical Interference and Noise	prone to Electrical interference and Noise.	Less prone to electrical Interference and Noise.
Data transfer Speed	Supports slower speed.	Supports higher speed.
Grounding	Not necessary	Necessary
Target Location	Homes & offices	Airports & factories

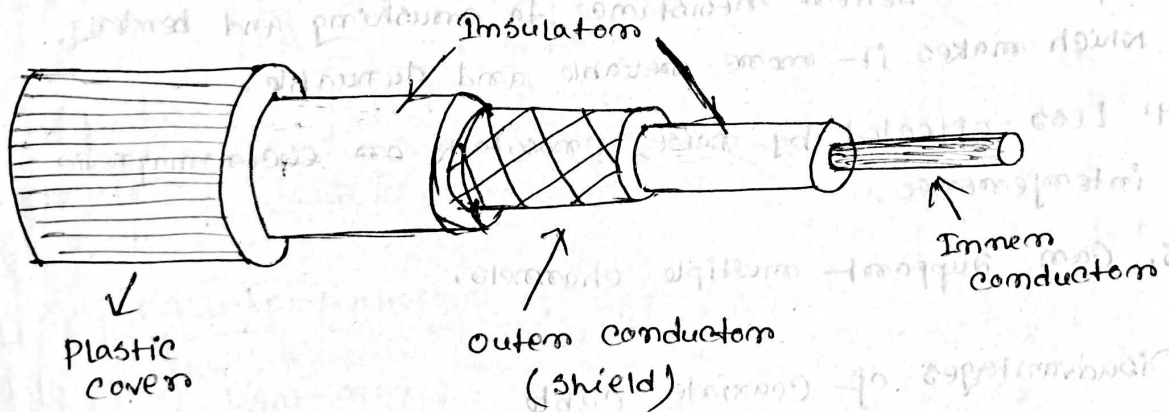


Coaxial Cable : A coaxial cable is an electrical cable that can carry a high volume of digital data in the form of signals, and the chances of signal loss in these cables are almost zero.

A coaxial cable consists of four basic parts —

- (i) Inner core conductor (the central conductor)
- (ii) Dielectric, which separates the inner and outer conductors.
- (iii) Outer conductor (shield)
- (iv) Jacket, which is the outermost polymer layer of the cable and protects the parts inside.

Having a copper conductor and strong dielectric insulator, surrounded by conductive shielding and a protective outer jacket, coaxial cables can prevent signal interference and crosstalk. This feature makes this cable best for video and audio devices.



Coaxial cable uses :

- (i) Coaxial cables are known for their reliable and accurate transmission. They are used in many homes for audio and visual purposes. These cables can be found connected from the wall outlet to a person's television or a set-top box.
- (ii) Coaxial cables are mainly used by phone companies, cable operators and internet service providers to transmit data, video and voice communications to homes. They are also increasingly used in local networking.

Coaxial cable types :

- There are two main types of coaxial cables — one with an impedance of 75 ohm(Ω) and another with an impedance of 50 ohm(Ω). ohm refers to the impedance. It is basically the measure of resistance in the cable to the flow of electrical energy. There is no good or bad impedance, only the right impedance for your application.
- Cables with 75 ohm are generally used for video signals, while 50 ohm cables are used for data and wireless communications.

Advantage of coaxial cable :

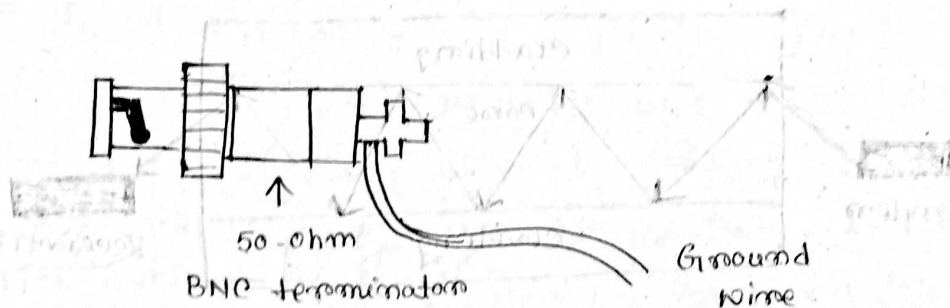
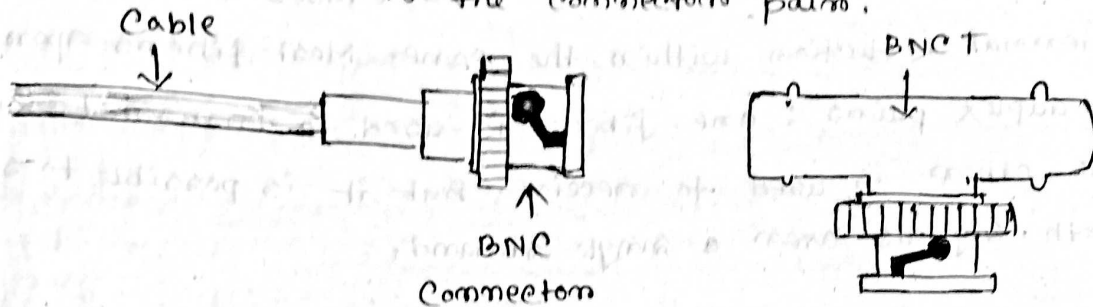
1. Supports high bandwidth levels.
2. Easy to install with simple tools.
3. provides better resistance to crushing and bending, which makes it more reliable and durable.
4. Less affected by noise, crosstalk or electromagnetic interference.
5. Can support multiple channels.

Disadvantages of Coaxial cable :

1. Expensive to install for longer distances.
2. Needs to be grounded to prevent interference.
3. Have multiple layers, making coaxial cable bulky.
4. Safety is an important concern since there is a risk of breaking the coaxial cable and attaching a "t-joint" by hackers.
5. Cable failure can cause your entire network to go down.

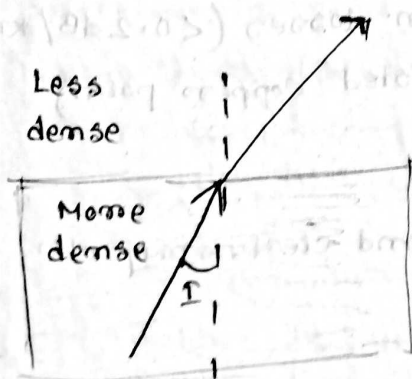
BNC Connectors :

BNC connectors are miniature radio frequency (RF) connectors developed from low-frequency applications. This quick-connect/disconnect circular connector features a two-stud bayonet-style locking mechanism that uses twist and snap into position. A tactile "snap" indicates a secure connection, a reverse motion disconnects the connector pair.

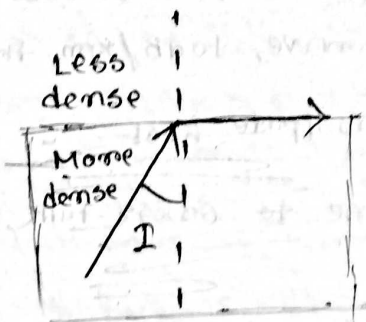


Coaxial Cable performance : PPT (5)

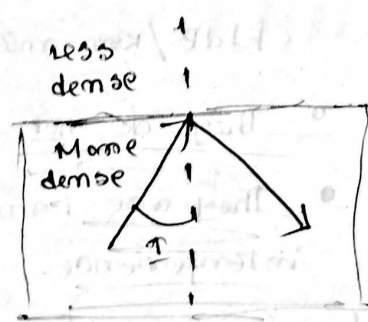
Bending of light ray :



$i < \text{critical angle}$,
refraction



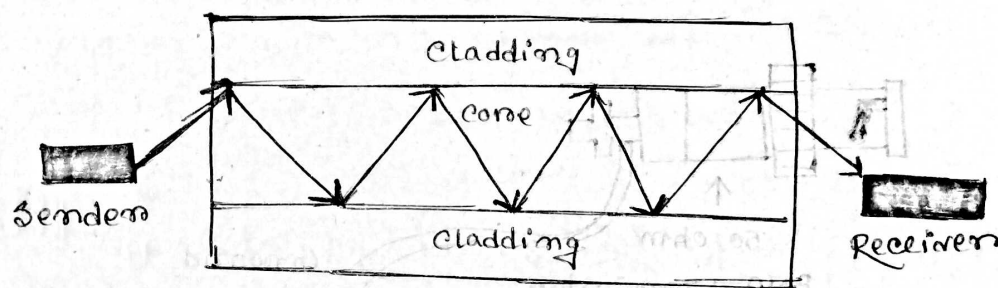
$i = \text{critical angle}$,
refraction



$i > \text{critical angle}$,
reflection

Optical fiber : Optical fiber is a very thin strand of pure glass which acts as a waveguide for light over long distances. It uses a principle known as total internal reflection.

Fiber optic cable is actually composed of two layers of glass : The core, which carries the actual light signal, and the cladding, which is a layer of glass surrounding the core. The cladding has a lower refractive index than the core. This causes ~~for~~ Total Internal Reflection within the core. Most fibers operate in duplex pairs : one fiber is used to transmit and the other is used to receive. But it is possible to send both signals over a single strand.



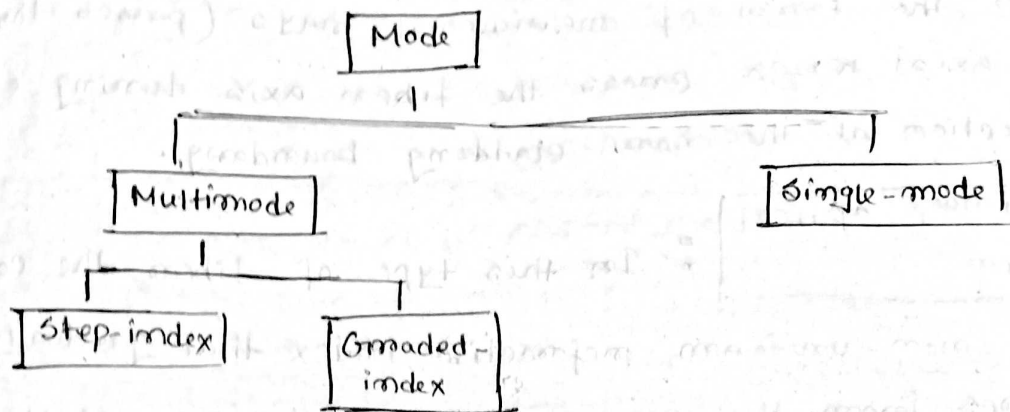
Why use optic fiber cable ?

- ⇒
- They have practically unlimited information.
 - They have high carrying capacity (very broad bandwidth, THz or Tbits/s)
 - They have very low transmission losses ($< 0.2 \text{ dB/km}$, cf 1 dB/km microwave, 10 dB/km twisted copper pair)
 - They do not dissipate heat.
 - They are immune to cross-talk and electromagnetic interference.

Types of Optical Fibers :

Based on RI profile and core size and Modes of propagation optical fibers are classified into 3 types —

- (1) Single mode step index fiber.
- (2) Multi mode step index fiber.
- (3) Graded index multi mode or Graded index fiber.



Single-mode fiber : In single-mode fiber, only one type of ray of light can propagate through the fiber. This type of fiber has a small core diameter (5 μm) and high cladding diameter (40 μm) and the difference between the refractive index of core and cladding is very small. There is no dispersion i.e. no degradation of the signal during traveling through the fiber. The light is passed through it through a laser diode.

Applications : telecoms.

Multi-mode fiber : Multimode fiber allows a large number of modes for the light traveling through it. The core diameter is generally (40 μm) and that of cladding is (70 μm). The relative refractive index difference is also greater than single mode fiber. There is signal degradation due to multimode dispersion. It is not suitable for long-distance communication due to large dispersion and attenuation of the signal. There are two categories on the basis of Multimode fiber i.e. Step Index Fiber and Graded Index Fiber. Basically these are categories of optical fiber on the basis of Refractive Index.

Applications : LANs

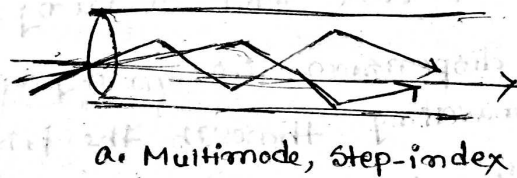
Step-index optical fiber

The refractive index of core is constant. The refractive index of the cladding is also constant. The rays of light propagate through it in the form of meridional rays (passes through the axis) which cross the fiber axis during every reflection at the core-cladding boundary.

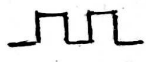
Graded-index optical fiber

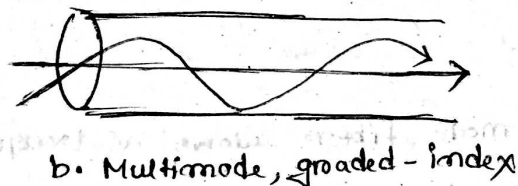
In this type of fiber, the core has a non-uniform refractive index that gradually decreases from the center towards the core-cladding interface. The cladding has a uniform refractive index. The light rays propagate through it in the form of skew rays or helical rays. It does not cross the fiber axis at any time.


Source

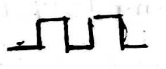


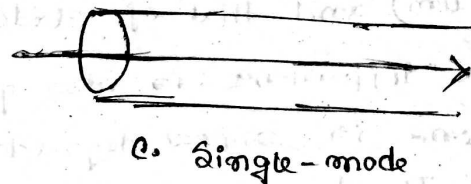

Destination


Source




Destination


Source




Destination

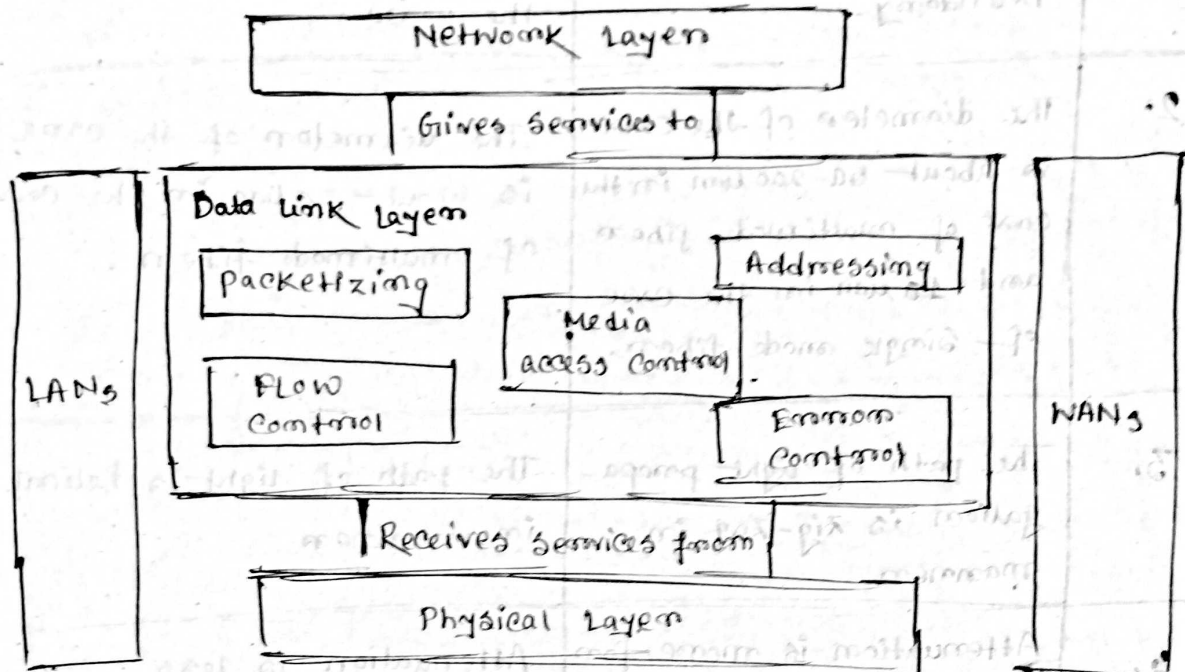
Sl No.	STEP INDEX FIBER	GRADED INDEX FIBER
1.	The refractive index of the core is uniform throughout and undergoes an abrupt change at the core cladding boundary.	The refractive index of the core is made to vary gradually such that it is maximum at the center of the core.
2.	The diameter of the core is about 50-200 μm in the case of multimode fibers and 10 μm in the case of single mode fibers.	The diameter of the core is about 50 μm in the case of multimode fibers.
3.	The path of light propagation is zig-zag in manner.	The path of light is helical in manner.
4.	Attenuation is more for multimode step index fiber but for single mode it is very less. Explanation: When a ray travels through the longer distances there will be some difference in reflected angles. Hence high angle rays arrive later than low angle rays causing dispersion resulting in distorted output.	Attenuation is less. Explanation: Here the light rays travel with different velocity in different paths because of their variation in their refractive indices. At the outer edge it travels faster than near the center. But almost all the rays reach the exit at the same time due to helical path. Thus, there is no dispersion.

→ SEE PPT (5)

PPT (6)

Data Link Layer

Position of the data-link layer:



- It is used for transferring the data from one node to another node.
- It receives the data from the network layer and converts the data into data frames and then attaches the physical address to these frames which are sent to the physical layer.
- It enables the error-free transfer of data from one node to another node.

Duties of Data Link Layer:

Framing: Data-link layer takes ~~pack~~ packets from Network layer and encapsulates them into frames. Then, it sends each frame bit-by-bit on the hardware. At receiver's end, data link layer picks up signals from hardware and assembles them into frames.

Addressing : Data-link layer provides layer-2 hardware addressing mechanism. Hardware address is assumed to be unique on the link. It is encoded into hardware at the time of manufacturing.

Synchronization : When data frames are sent on the link, both machines must be synchronized in order to transfer to take place.

Error Control : Sometimes signals may have encountered problem in transition and the bits are flipped. These errors are detected and attempted to recover actual data bits. It also provides error reporting mechanism to the sender.

Flow Control : Stations on same link may have different speed or capacity. Data-link layer ensures a flow control that enables both machine to exchange data on same speed.

Multi-Access : When host on the shared link tries to transfer the data, it has a high probability of collision. Data-link layer provides mechanism such as CSMA/CD to equip capability of accessing a shared media among multiple system.

LLC and MAC sublayers
IEEE standards for LANs } PPT 6

From Error Detection and connection

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